A Process-Relational Foundation for Geometry

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Abstract

Classical geometry takes space and points as primitives, treating metric and positional properties as intrinsic to geometric objects. We argue that these are not intrinsic but relational: they arise from explicit comparisons with chosen reference structures. Accordingly, we develop a foundational framework in which only finite relations and their compositions are primitive. Finite composable sequences of relations, called *configurations*, represent processes. Objects arise as equivalence classes under a *congruence*—an equivalence relation compatible with composition—that expresses observational indistinguishability.

Our main results are:

- **Theorem 5.2 (Quotient Construction).** Any congruence on an associative relational system yields a well-defined quotient with associative composition.
- **Theorem 6.3 (Metric Emergence).** A weight function on a relational system with inverses induces a pseudometric—and under a separation condition, a genuine metric—on the quotient, derived from path optimization. When weights are integer-valued and R is finite, the infimum is a minimum (Corollary 6.4).
- **Theorem 7.1 (Reconstruction).** Every finite metric space arises, up to isometry, from a relational system. The framework is universal for finite metric geometry.
- **Theorem 11.10 (Categorical Equivalence).** After identifying metrically equivalent relational systems, the metric realization functor yields an equivalence of categories between relational systems and finite metric spaces.

Concrete discrete constructions of the circle ($\mathbb{Z}/n\mathbb{Z}$), torus, tree (negative curvature), and triangulated sphere (positive curvature) illustrate the framework, along with a discrete relational account of curvature via noncommutative composition. A dedicated section establishes compatibility with classical geometry via Gromov–Hausdorff convergence, framed as a bridge result rather than a foundational claim. A further section establishes that space itself is not a primitive container but emerges from inter-system relational structure: when two relational systems are connected by weighted relations, the combined system induces a finite metric space whose geometry is fully determined by the internal structure of each object and the connecting relations between them, with position encoded entirely in those inter-system relations. No continuity, limits, infinitesimals, or infinite processes are assumed as foundational primitives; infinite discrete structures such as the regular tree arise as derived objects from the framework's finite rules, not as inputs.

1 Introduction

1.1 The Classical Approach and Its Hidden Assumptions

Standard geometry begins with a background space—typically Euclidean $\mathbb{R}^n$—populated by points, and defines all further structure (distance, angle, adjacency) on top of this substrate. This approach has been enormously productive, but it embeds several assumptions that are rarely made explicit.

First, *position* is treated as an intrinsic property of a geometric object, when in fact the position of any object is only meaningful relative to a coordinate system or reference frame. A triangle does not "have" a location; it has a location *relative to something else*.

Second, *metric structure* is similarly treated as ambient and pre-given, when it is in fact the result of a structural comparison between two objects: the geometric object being measured and the reference system

supplying the unit. The length of a segment is not a property of the segment alone, nor a property imposed by any observer. It is what emerges at the interface between the segment and its measurement context — produced by their contact, belonging to neither one separately.

Third, the classical framework takes *continuity and infinity* as primitive, encoding them in the real number substrate. Yet no physical measurement or finite computation ever directly accesses the continuum. These are idealizations, and treating them as primitive conflates the map with the territory.

1.2 A Process-Relational Alternative

We propose a foundational framework that takes none of these for granted. The only primitives are *finite relations* and their *compositions*. No background space, no coordinates, no points, no continuous structures, and no infinite processes are assumed. All geometric structure is derived from finite relational data.

The central move is this: *objects are not primitive entities but equivalence classes of finite relational processes*. A geometric object is not a point or a set of points placed in space; it is the entire class of processes that no relational test can distinguish. Space is not assumed; it emerges as the structured set of such classes.

The framework rests on four explicit commitments:

1. **Relations and compositions are primitive.** There is no substrate in which relations live; relations are the substrate.

2. **Objects are equivalence classes of processes.** A geometric object is not a point or a set of points, but an equivalence class of finite composable sequences of relations under a congruence. This defines what an object *is*.

3. **Congruence encodes observational identity.** The criterion by which two processes are identified is that no relational test distinguishes them: composing any third configuration on either side produces the same result. This defines when two processes are the *same* object.

4. **All constructions are finite and discrete.** Continuity and infinity are not part of the framework; they are not approximated, they are simply absent.

Commitment 4 is a foundational choice, not a limitation. Geometric structure should be expressible in terms of finite relational data, or not claimed to exist within this framework at all. The framework thereby enforces a discipline: every geometric claim must be grounded in explicit, finite, computable relational structure.

1.3 Motivation

Several independent developments across mathematics, physics, and computation suggest that a discrete relational foundation for geometry is not merely an intellectual exercise but a practical and physical necessity.

In *quantum gravity*, approaches including causal set theory [12], loop quantum gravity, and spin foam models proceed from the premise that spacetime at the Planck scale is fundamentally discrete and relational. The continuum is expected to be an emergent approximation, not a primitive substrate. These frameworks require a rigorous mathematical language for discrete relational geometry that does not presuppose the smooth manifolds it is trying to replace. The present framework offers one such language: whether it is the right language for any specific quantum gravity model is an open question, but it demonstrates that rigorous discrete relational geometry is mathematically feasible without a continuous substrate.

In *computational geometry and digital topology*, geometric structures are always represented as finite discrete data. Classical geometry must be discretized for computational use, introducing approximation errors that must be managed. A framework that is natively discrete requires no approximation: its objects are already finite, its operations are already exact, and its equivalences are already computable. The framework offers a foundational account of why discrete computational geometry is not a degraded approximation of the real thing but a fully legitimate geometric system in its own right.

In *mathematical foundations*, the specific primitives of classical geometry — the point, the infinite line, the continuous circle — are not logically inevitable. They reflect the cognitive and historical conditions

under which geometry was first formalized: the experience of space as pre-existing, of measurement as always available, of the continuum as the natural habitat of number. A framework that derives geometric objects from finite relational data, treating metric, position, and scale as external contextual parameters rather than intrinsic primitives, is not rejecting classical geometry. It is identifying which of its features are necessary and which are contingent on the conditions of its historical development.

In summary, the relational approach offers three principal advantages. First, *conceptual clarity*: by separating what is intrinsic to a geometric object from what is supplied by the external context of observation, the framework eliminates a class of conflations that have persisted in classical foundations. Second, *computational tractability*: because all objects are finite and all operations are exact, every geometric question reduces to a finite combinatorial computation without approximation. Third, *alignment with discrete physical theories*: the framework provides a rigorous mathematical language for spacetime models that treat discreteness and relationality as primitive, rather than approximating them from a continuous substrate.

1.4 Main Contributions and Paper Structure

The central results are the Quotient Construction (Theorem 5.2), Metric Emergence (Theorem 6.3), Reconstruction of Finite Metric Spaces (Theorem 7.1), and the Categorical Equivalence (Theorem 11.10).

Section 2 defines relational systems and their directed multigraph presentation. Section 3 introduces configurations as finite processes. Section 4 defines congruence. Section 5 proves the quotient theorem and establishes a complete classification of single-generator quotients, including a partial classification of congruences. Section 6 develops metric emergence in full generality, with a corollary for the discrete case. Section 7 establishes metric universality, minimal presentations, failure modes, and the domain of validity. Section 8 provides concrete constructions including a full end-to-end worked example and negative, zero, and positive curvature examples. Section 9 and Section 10 connect the framework to category theory and higher structures. Section 11 develops the enriched categorical formulation and the equivalence theorem. Section 13 establishes compatibility with classical geometry. Section 14 develops multi-object configurations and the emergence of space from inter-system relations. Section 12 situates the work relative to existing frameworks, including an explicit statement of what is new versus reformulated. Section 15 summarizes and suggests directions.

Notation. Throughout, R denotes a set of primitive relations, $\circ$ a partially defined composition, $\mathcal{C}$ the set of all configurations, and $\sim$ a congruence on $\mathcal{C}$. We write $[C]$ for the equivalence class of a configuration C, and $\mathcal{C}/\!\sim$ for the quotient. The empty configuration is denoted id. The symbol $\sim$ is used consistently for a congruence; observational indistinguishability is a *property* relative to a given $\sim$, not a second relation (see Definition 4.2 and Remark 4.3).

2 Relational Systems

Definition 2.1 (Relational System)**.** A *relational system* is a pair $(R, \circ)$ where:

- R is a non-empty set of *primitive relations*,
- $\circ : R \times R \rightharpoonup R$ is a *partially defined* composition.

Definition 2.2 (Compatibility)**.** Relations $r_1, r_2 \in R$ are *compatible* if $r_1 \circ r_2$ is defined.

Definition 2.3 (Associativity)**.** A relational system is *associative* if, whenever all compositions involved are defined,

$$(r_1 \circ r_2) \circ r_3 = r_1 \circ (r_2 \circ r_3).$$

All relational systems in this paper are assumed associative.

Definition 2.4 (Relational Inverse)**.** A relation $r \in R$ has an *inverse* $r^{-1} \in R$ if

$$r \circ r^{-1} = \mathrm{id} \quad \text{and} \quad r^{-1} \circ r = \mathrm{id}.$$

A relational system is *invertible* if every relation has an inverse. In a system with congruence $\sim$, we say r has an *inverse up to* $\sim$ if $r \circ r^{-1} \sim \mathrm{id}$ and $r^{-1} \circ r \sim \mathrm{id}$.

Remark 2.5 (Partial composition). Partial composition is essential: requiring all pairs to compose would force a uniformity inconsistent with most geometric and relational structures of interest. The partial setting allows the framework to encode directedness, type constraints, and locality naturally.

Remark 2.6 (Directed multigraph presentation). A relational system $(R, \circ)$ admits an equivalent formulation as a *weighted directed multigraph* $G = (V, E, s, t, w)$, where V is a set of vertices, E a set of edges, $s, t : E \to V$ assign source and target to each edge, and $w : E \to [0, \infty)$ is a weight function. Primitive relations correspond to edges; compatibility of r_1 and r_2 corresponds to $t(r_1) = s(r_2)$; composition corresponds to concatenation of composable edges.

This formulation makes domain and codomain tracking explicit and clarifies that the underlying structure is a *free category on a directed graph*, quotiented by the congruence $\sim$. In Section 7 and Section 11 we use this presentation to state the universality and categorical equivalence results precisely.

3 Configurations as Finite Processes

Definition 3.1 (Configuration)**.** A *configuration* is a finite composable sequence of relations:

$$C = (r_1, r_2, \ldots, r_n), \quad r_i \in R,$$

such that each consecutive pair r_i, r_{i+1} is compatible (Definition 2.2). The *empty configuration* id (of length 0) is included as a two-sided identity. The set of all configurations is denoted $\mathcal{C}$.

The term *configuration* is used uniformly throughout. Configurations represent finite, discrete processes or traversals in the relational system. Informally, a configuration is a finite sequence of relational steps that can be composed end-to-end.

Definition 3.2 (Configuration Composition)**.** For $C = (r_1, \ldots, r_m)$ and $D = (s_1, \ldots, s_n)$, their *composition* is:

$$C \circ D = (r_1, \ldots, r_m, s_1, \ldots, s_n),$$

defined when r_m and s_1 are compatible. We have $\mathrm{id} \circ C = C \circ \mathrm{id} = C$ for all $C \in \mathcal{C}$.

Remark 3.3 (Finiteness of configurations). No limit, no infinite sequence, and no continuity condition is involved. Every configuration is a finite object. Composition is concatenation, a purely discrete, computable operation.

4 Congruence and Observational Indistinguishability

Definition 4.1 (Congruence)**.** An equivalence relation $\sim$ on $\mathcal{C}$ is a *congruence* (with respect to composition) if, for all configurations $C_1, C_1', C_2, C_2' \in \mathcal{C}$,

$$C_1 \sim C_1' \text{ and } C_2 \sim C_2' \;\Rightarrow\; C_1 \circ C_2 \sim C_1' \circ C_2',$$

whenever the compositions are defined.

A congruence is the appropriate notion of equivalence for a system with composition: it says that substituting equivalent configurations preserves equivalence of the result. This is the only property of $\sim$ needed for Theorem 5.2.

Definition 4.2 (Observational Indistinguishability)**.** Given a congruence $\sim$ on $\mathcal{C}$, configurations C_1 and C_2 are *observationally indistinguishable (with respect to $\sim$)* if, for every $D \in \mathcal{C}$ compatible with both,

$$C_1 \circ D \sim C_2 \circ D \quad \text{and} \quad D \circ C_1 \sim D \circ C_2.$$

Remark 4.3 (No circularity). Definition 4.2 does not define $\sim$; it describes a property that a pair (C_1, C_2) may or may not have *relative to a given* congruence $\sim$. The intended workflow is:

1. Fix a relational system $(R, \circ)$.

2. Specify $\sim$ by listing the relational laws of the system (e.g., $a^n \sim \mathrm{id}$).
3. Verify that $\sim$ satisfies Definition 4.1.
4. Observe as a consequence that equivalent configurations are observationally indistinguishable.

In practice, step 4 follows from steps 1–3 by the congruence property. Observational indistinguishability is thus a *derived* property, not a primitive one.

5 Objects and the Quotient Construction

Definition 5.1 (Object)**.** An *object* is an equivalence class $[C]$ of a configuration C under a congruence $\sim$:

$$[C] = \{C' \in \mathcal{C} \mid C' \sim C\}.$$

Objects are the geometric entities of the framework. A single geometric "thing" is not a configuration but the entire class of all configurations that no *relational test* can distinguish from one another.

Theorem 5.2 (Quotient Construction)**.** *Let $(R, \circ)$ be an associative relational system with configuration set $\mathcal{C}$. Let $\sim$ be a congruence on $\mathcal{C}$. Then:*

(i) The quotient $\mathcal{C}/\!\sim$ is well-defined as a set of equivalence classes.

(ii) Composition descends to a well-defined operation on $\mathcal{C}/\!\sim$:

$$[C_1] \circ [C_2] \;:=\; [C_1 \circ C_2].$$

(iii) The induced composition is associative.

(iv) The class $[\mathrm{id}]$ *acts as a two-sided identity in $\mathcal{C}/\!\sim$.*

Proof. **(i).** By definition, $\sim$ partitions $\mathcal{C}$ into disjoint, non-empty equivalence classes.

(ii). We verify independence of representatives. Suppose $[C_1] = [C_1']$ and $[C_2] = [C_2']$, i.e., $C_1 \sim C_1'$ and $C_2 \sim C_2'$. Since $\sim$ is a congruence (Definition 4.1),

$$C_1 \circ C_2 \sim C_1' \circ C_2',$$

hence $[C_1 \circ C_2] = [C_1' \circ C_2']$. The operation is independent of the choice of representatives.

(iii). For $C_1, C_2, C_3 \in \mathcal{C}$, associativity at the configuration level gives $(C_1 \circ C_2) \circ C_3 = C_1 \circ (C_2 \circ C_3)$. Applying part (ii):

$$([C_1] \circ [C_2]) \circ [C_3] = [(C_1 \circ C_2) \circ C_3] = [C_1 \circ (C_2 \circ C_3)] = [C_1] \circ ([C_2] \circ [C_3]).$$

(iv). For any $C \in \mathcal{C}$, $\mathrm{id} \circ C = C \circ \mathrm{id} = C$, hence $[\mathrm{id}] \circ [C] = [C] = [C] \circ [\mathrm{id}]$. □

Remark 5.3 (The quotient reveals, not creates)*.* The *quotient* does not introduce structure. It stabilizes invariants already present in $(R, \circ)$ by collapsing configurations that are indistinguishable within the relational system. Geometric structure is encoded in the generators R and laws $\sim$; the quotient merely makes that structure manifest.

5.1 Taxonomy of Single-Generator Quotients

The quotient construction produces different geometric objects depending on the congruence. For single-generator systems, the possibilities are completely classified. This classification is foundational: it characterizes the expressive range of the simplest relational systems and provides the algebraic criterion used throughout the examples of Section 8.

Proposition 5.4 (Conditions for a well-formed closed object)**.** *Let $(R, \circ)$ be an associative relational system with a single generator a and congruence $\sim$. The quotient $\mathcal{C}/\!\sim$ is a well-formed closed object if and only if the following two conditions hold simultaneously:*

(i) ***Closure.*** *There exists $n \geq 1$ such that $a^n \sim \mathrm{id}$.*

(ii) ***Uniqueness.*** *All intermediate steps are distinct: $a^j \not\sim a^k$ for all $0 \leq j < k < n$.*

When both hold, the quotient is $\mathbb{Z}/n\mathbb{Z}$. When closure holds but uniqueness fails, the shape collapses to a degenerate cycle. When closure fails entirely, the shape is open.

Proof. If $a^n \sim \mathrm{id}$ and all $a^0, \ldots, a^{n-1}$ are pairwise inequivalent, the n classes $[a^0], \ldots, [a^{n-1}]$ are distinct and $\mathcal{C}/\!\sim \cong \mathbb{Z}/n\mathbb{Z}$ by the congruence property. If $a^j \sim a^k$ for some $j < k < n$, the quotient has period $k - j < n$. If no n satisfies $a^n \sim \mathrm{id}$, all powers are distinct and the shape is open. □

Lemma 5.5 (Reduction to minimal period)**.** *Let $(R, \circ)$ be an associative relational system with a single generator a and congruence $\sim$. If there exists $n \geq 1$ such that $a^n \sim \mathrm{id}$ but the intermediate composites $a^0, a^1, \ldots, a^{n-1}$ are not all pairwise inequivalent, then there exists a strictly smaller $k < n$ such that $a^k \sim \mathrm{id}$.*

Proof. If intermediate composites are not all distinct, there exist indices $i < j \leq n$ such that $a^i \sim a^j$. By the congruence property, composing both sides on the right with a^{-i} gives

$$\mathrm{id} \;\sim\; a^i \circ a^{-i} \;\sim\; a^j \circ a^{-i} \;=\; a^{j-i}.$$

So $a^{j-i} \sim \mathrm{id}$ with $k := j - i < n$. Repeating this reduction if necessary produces a minimal-period representative in which all intermediate composites $a^0, a^1, \ldots, a^{k-1}$ are pairwise inequivalent. □

Proposition 5.6 (Taxonomy of closure)**.** *Let $(R, \circ)$ be an associative relational system with a single generator a and congruence $\sim$. By Lemma 5.5, it suffices to classify systems at their minimal period, i.e., under the assumption that all intermediate composites are distinct. Exactly one of the following holds for such minimal-period systems:*

(i) ***Closed.*** *$a^n \sim \mathrm{id}$ for some $n \geq 1$ with all intermediate steps distinct. Quotient: $\mathbb{Z}/n\mathbb{Z}$.*

(ii) ***Pseudo-periodic.*** *$a^n \sim \phi$ for some $n \geq 1$ and $\phi \not\sim \mathrm{id}$, with all intermediate steps distinct.*

(iii) ***Open.*** *No periodicity law holds. Quotient: $\mathbb{Z}$.*

Proof. By Lemma 5.5, any system with repeated intermediate composites reduces to one with strictly smaller period. It therefore suffices to classify minimal-period systems.

For such systems, cases (i) and (ii) are distinguished by whether the return target is id or some other $\phi \not\sim \mathrm{id}$. A system cannot belong to both: if $a^n \sim \mathrm{id}$ then by definition the return target is id, not a nontrivial ϕ.

Case (iii) covers all systems for which no finite composition returns to any previously visited class; in this case all powers $a^0, a^1, \ldots$ are distinct. This is the negation of cases (i) and (ii), so the three cases are mutually exclusive and exhaustive. □

5.2 A Partial Classification of Congruences

The paper's central claim is that geometry is determined by the choice of congruence laws. It is therefore natural to ask: when do two different congruences on the same relational system produce the same geometric object? We give a complete answer for the single-generator case.

We first define the key invariants intrinsically in relational language, with no appeal to external symmetry notions.

Definition 5.7 (Intrinsic period and shift order)**.** Let $R = \{a, a^{-1}\}$ and let $\sim$ be a congruence on $\mathcal{C}$.

(i) The *period* $n \in \mathbb{N} \cup \{\infty\}$ is the minimal $n > 0$ such that $a^n \sim \mathrm{id}$, if it exists, and ∞ otherwise.

(ii) When $n < \infty$, define the *loop subgroup*

$$S \;:=\; \big\{[m] \in \mathbb{Z}/n\mathbb{Z} \;\big|\; a^m \sim \mathrm{id}\big\} \;\subseteq\; \mathbb{Z}/n\mathbb{Z}.$$

The *shift order* is $k := n/|S|$.

Lemma 5.8. *The set S in Definition 5.7 is a subgroup of $\mathbb{Z}/n\mathbb{Z}$.*

Proof. If $a^m \sim \mathrm{id}$ and $a^p \sim \mathrm{id}$, then by the congruence property $a^{m+p} = a^m \circ a^p \sim \mathrm{id} \circ \mathrm{id} \sim \mathrm{id}$, so $[m+p] \in S$. If $a^m \sim \mathrm{id}$, then composing both sides on the right by a^{-m} gives $a^m \circ a^{-m} \sim \mathrm{id} \circ a^{-m}$, i.e., $\mathrm{id} \sim a^{-m}$, i.e., $a^{-m} \sim \mathrm{id}$, so $[-m] \in S$. Hence S is closed under addition and negation in $\mathbb{Z}/n\mathbb{Z}$. □

Remark 5.9 (Interpretation of k). When $S = \{[0]\}$, no nontrivial power of a is identified with id within one period: the system is a pure cycle and $k = n$. The case $S = \mathbb{Z}/n\mathbb{Z}$ — where every power of a would be identified with id — is unreachable under the minimal-period assumption of Proposition 5.6: by Lemma 5.5, any such system reduces to period 1. The relevant range is therefore $\{[0]\} \subseteq S \subsetneq \mathbb{Z}/n\mathbb{Z}$, equivalently $1 < k \leq n$, with $k = n$ corresponding to the closed (pure cycle) case and $1 < k < n$ to the pseudo-periodic cases. Intermediate subgroups yield intermediate values of k, encoding the internal loop structure of the orbit. The shift order k is the index $[\mathbb{Z}/n\mathbb{Z} : S]$.

Proposition 5.10 (Complete congruence classification for single-generator systems)**.** *Let $R = \{a, a^{-1}\}$ be the single-generator invertible system, and let $\sim_1$, $\sim_2$ be congruences on $\mathcal{C}$ with associated invariants (n_1, k_1) and (n_2, k_2) as in Definition 5.7. The quotients $\mathcal{C}/\!\sim_1$ and $\mathcal{C}/\!\sim_2$ are isomorphic if and only if one of the following holds:*

(i) $n_1 = n_2 = \infty$ (both open; both quotients are isomorphic to $\mathbb{Z}$);

(ii) $n_1, n_2 < \infty$, $k_1 = k_2 = 1$, and $n_1 = n_2$ (both closed with the same period);

(iii) $n_1, n_2 < \infty$, $k_1, k_2 > 1$, and $n_1 = n_2$ and $k_1 = k_2$ (both pseudo-periodic with the same period and shift order);

(iv) No other combination yields isomorphic quotients.

Proof. **Cases (i) and (ii).** Open congruences yield $\mathbb{Z}$; closed congruences with period n yield $\mathbb{Z}/n\mathbb{Z}$. These are isomorphic as monoids if and only if the periods agree ($n_1 = n_2$). A finite monoid is never isomorphic to $\mathbb{Z}$.

Case (iii). We prove both directions.

($\Rightarrow$) Isomorphic quotients imply equal (n, k). If $\Phi : \mathcal{C}/\!\sim_1 \to \mathcal{C}/\!\sim_2$ is an isomorphism, it must send the generator $[a]_1$ to a generator of $\mathcal{C}/\!\sim_2$. Since $\mathcal{C}/\!\sim_2$ is generated by $[a]_2$, the isomorphism preserves the period: $a^{n_1} \sim_1 \mathrm{id}$ implies $\Phi([a]_1)^{n_1} = \Phi([a^{n_1}]_1) = \Phi([\mathrm{id}]_1) = [\mathrm{id}]_2$, so $a^{n_1} \sim_2 \mathrm{id}$, giving $n_2 \leq n_1$. By symmetry $n_1 \leq n_2$, hence $n_1 = n_2 =: n$. Similarly, an isomorphism maps the loop subgroup S_1 bijectively to S_2 (since it preserves the identity and the group operation), so $|S_1| = |S_2|$ and thus $k_1 = n/|S_1| = n/|S_2| = k_2$.

($\Leftarrow$) Equal (n, k) implies isomorphic quotients. Assume $n_1 = n_2 = n$ and $k_1 = k_2$. Since $k = n/|S|$ and $S \leq \mathbb{Z}/n\mathbb{Z}$, equal k with equal n implies $|S_1| = |S_2|$. Since $\mathbb{Z}/n\mathbb{Z}$ is cyclic, it has exactly one subgroup of each order dividing n; therefore $S_1 = S_2 =: S$.

Define the map $\Phi : \mathcal{C}/\!\sim_1 \to \mathcal{C}/\!\sim_2$ by $\Phi([a^j]_1) := [a^j]_2$.

Well-definedness. Suppose $[a^j]_1 = [a^{j'}]_1$, i.e., $a^j \sim_1 a^{j'}$. By the congruence property, $a^{j-j'} \sim_1 \mathrm{id}$, so $[j - j'] \in S_1 = S_2$, hence $a^{j-j'} \sim_2 \mathrm{id}$, giving $a^j \sim_2 a^{j'}$ and $[a^j]_2 = [a^{j'}]_2$.

Bijectivity. The map $\Psi([a^j]_2) := [a^j]_1$ is the inverse by the same argument with $\sim_1$ and $\sim_2$ swapped.

Composition-preservation. $\Phi([a^j]_1 \circ [a^m]_1) = \Phi([a^{j+m}]_1) = [a^{j+m}]_2 = [a^j]_2 \circ [a^m]_2 = \Phi([a^j]_1) \circ \Phi([a^m]_1)$.

Hence Φ is an isomorphism and $\mathcal{C}/\!\sim_1 \cong \mathcal{C}/\!\sim_2$.

Case (iv). Across types, closed quotients (finite, no nontrivial internal loops) are never isomorphic to pseudo-periodic quotients (finite, with nontrivial S), nor to open quotients (infinite). □

Remark 5.11 (The invariant pair (n, k) as a geometric signature). The pair (n, k) is a complete isomorphism invariant for single-generator quotients: it determines the quotient up to isomorphism and is determined by it. It encodes two independent pieces of geometric information — n measures the scale of the orbit (how many steps before any return), and k measures the internal symmetry (how much loop structure is present within that orbit). For multi-generator systems the analogous invariant is a normal subgroupoid of the free path groupoid, which generalizes both n and k. A complete invariant theory for that case is the central open problem of this framework.

Remark 5.12 (The multi-generator classification gap). The single-generator classification of Proposition 5.10 is complete: any two congruences on $R = \{a, a^{-1}\}$ are classified by the pair (n, k). For multi-generator systems the situation is substantially more complex, and no analogous complete classification is currently known within the framework.

What is known is the following. The framework is *universal*: by Theorem 7.1 and the categorical equivalence of Theorem 11.10, every finite metric space arises from some multi-generator relational system. Universality means any geometry can be represented. It does not mean the space of possible geometries is understood or classified.

The gap is therefore not a gap in expressiveness but a gap in understanding. Classical geometry faces the same situation: every Riemannian manifold can be described, but the classification of all Riemannian manifolds up to isometry is an open problem in differential geometry. The relational framework recovers this difficulty in its own terms: describing all possible congruences on a multi-generator free path groupoid, and determining when two produce isomorphic quotients, is the relational version of the same question. The natural invariant is the normal subgroupoid structure of the kernel, which generalizes the subgroup $S \leq \mathbb{Z}/n\mathbb{Z}$ of the single-generator case to the groupoid setting. Developing this into a complete invariant theory is the framework's central open problem.

6 Metric Emergence

We show how a notion of distance arises from the relational system without being imposed as a primitive. We state the general theorem for real-valued weights first, then specialize to the integer-valued case used throughout the finite discrete framework.

Definition 6.1 (Weight Function)**.** A *weight function* is a map $w : R \to [0, \infty)$, extended to configurations by:

$$w(C) = w(r_1) + \cdots + w(r_n) \quad \text{for } C = (r_1, \ldots, r_n), \qquad w(\mathrm{id}) = 0.$$

When w takes values in $\mathbb{N}$, we call it a *discrete weight function.*

Definition 6.2 (Relational Path Between Objects)**.** A *relational path from* $[A]$ *to* $[B]$ is a configuration $P \in \mathcal{C}$ such that, for some representative $a \in [A]$, the configuration $a \circ P$ belongs to $[B]$. The set of all such paths is denoted $\mathrm{Path}([A], [B])$.

Theorem 6.3 (Metric Emergence)**.** *Let $(R, \circ)$ be an invertible, associative relational system with congruence $\sim$ and weight function $w : R \to [0, \infty)$. Assume:*

(A1) $w(r) \geq 0$ *for all* $r \in R$*,*

(A2) $w(C_1) = w(C_2)$ *whenever* $C_1 \sim C_2$ *(weight is a congruence invariant),*

(A3) $w(\mathrm{id}) = 0$ *(the identity has zero cost),*

(A4) $w(r^{-1}) = w(r)$ *for all* $r \in R$ *(inverses have equal weight).*

Define:

$$d([A], [B]) = \inf\big\{w(P) \;\big|\; P \in \mathrm{Path}([A], [B])\big\}.$$

Then d is a pseudometric *on $\mathcal{C}/\!\sim$. If additionally:*

(A5) $d([A], [B]) = 0 \Rightarrow [A] = [B]$ *(separation),*

then d is a metric*.*

Proof. **Non-negativity.** Immediate from (A1) and the definition as an infimum over non-negative values.

Self-distance. $d([A], [A]) = 0$ because $\mathrm{id} \in \mathrm{Path}([A], [A])$ and $w(\mathrm{id}) = 0$ by (A3).

Symmetry. By invertibility and (A4), reversing a path $P = (r_1, \ldots, r_n)$ from $[A]$ to $[B]$ yields $P^{-1} = (r_n^{-1}, \ldots, r_1^{-1})$, a path from $[B]$ to $[A]$ with $w(P^{-1}) = w(P)$. Hence $d([A], [B]) = d([B], [A])$.

Triangle inequality. Given $P \in \mathrm{Path}([A],[B])$ and $Q \in \mathrm{Path}([B],[C])$, the concatenation $P \circ Q \in \mathrm{Path}([A],[C])$ satisfies $w(P \circ Q) = w(P) + w(Q)$ by additivity of w. Taking infima:

$$d([A],[C]) \leq d([A],[B]) + d([B],[C]).$$

Metric. Assumption (A5) upgrades the pseudometric to a metric by providing separation. □

Corollary 6.4 (Discrete metric emergence)**.** *If additionally $w : R \to \mathbb{N}$ is a discrete weight function and R is finite with $w(r) \geq 1$ for all $r \in R$, then the infimum in Theorem 6.3 is attained:*

$$d([A],[B]) = \min\{w(P) \mid P \in \mathrm{Path}([A],[B])\}.$$

No limiting process or appeal to completeness is required.

Proof. When $w(r) \geq 1$ for all r, any path of k steps has weight at least k. For a fixed target weight M, the set of paths with $w(P) \leq M$ is a finite set (bounded length, finite R). The infimum over a finite non-empty set of non-negative integers is a minimum. □

Remark 6.5 (Scope of assumption (A5))*.* Assumption (A5) is not automatically satisfied. If w assigns weight zero to some non-identity relation r with $r \not\sim \mathrm{id}$, then $d([A],[A \circ r]) = 0$ even though $[A] \neq [A \circ r]$, and d is a pseudometric but not a metric. The separation condition holds whenever every zero-weight configuration is equivalent to id under $\sim$. In particular, if $w(r) > 0$ for all $r \in R$, then any non-empty path has positive weight and separation holds automatically.

Remark 6.6 (Metric as ambient comparison, not observer-relative quantity)*.* The framework's treatment of metric as an external parameter might suggest that distances are observer-relative — that they depend on who is measuring. This reading should be resisted. A distance in this framework is not relative to a subject. It is the result of a *structural comparison* between two mathematical objects: the relational system being measured and the reference system supplying the weight function. Neither party in that comparison is a mind or a perspective. Both are mathematical structures.

A useful analogy is thermal equilibrium in physics. Temperature is not a property of a body alone, nor of the thermometer alone, nor of a physicist performing the measurement. It is a property of the contact between two physical systems at equilibrium. No subject is required. The quantity emerges from the interface between the two structures.

Length in this framework has the same character. A circle of resolution $n = 100$ and one of resolution $n = 1000$ are the same relational object. The different values of n are not two observers seeing the same thing differently — they are the results of two different structural comparisons, each involving a different reference step size. The length is neither intrinsic to the circle nor externally imposed by a subject. It is produced at the interface between the relational object and its measurement context.

This positions the framework at a point distinct from both classical realism (length is intrinsic) and constructivism or observationalism (length is what an observer measures). The correct description is *ambient comparison*: a quantity that belongs to the contact between two structures, is fully determined by that contact, and requires no subject to perform it. The resolution parameter is not a cognitive choice — it is the scale at which the measurement context operates. Changing it changes the comparison, not the object being compared.

7 Metric Universality

Theorem 6.3 shows that a relational system equipped with a weight function produces a metric. We now establish the converse: every finite metric space arises from a relational system. This demonstrates that the framework is not merely generative but *universal* for finite metric geometry. The reconstruction uses real-valued weights $w : R \to [0, \infty)$, the general case of Definition 6.1.

7.1 Reconstruction of Finite Metric Spaces

Theorem 7.1 (Reconstruction of finite metric spaces)**.** *Let (X, d) be a finite metric space. Then there exists a relational system $(R, \circ)$ with weight function $w : R \to [0, \infty)$ such that the induced metric on $R/\sim$ is isometric to (X, d).*

Proof. We construct the relational system explicitly.

Step 1: Construction. Set $R := \{e_{xy} \mid x, y \in X\}$, with e_{xy} representing a direct relation from x to y. Define composition by $e_{xy} \circ e_{yz} = e_{xz}$ whenever the target of e_{xy} matches the source of e_{yz}, i.e., when the middle vertex agrees. Assign weights $w(e_{xy}) := d(x, y)$. The identity at each x is e_{xx} with $w(e_{xx}) = d(x, x) = 0$.

Step 2: Induced distance dominates d. The single-edge configuration (e_{xy}) connects x to y with $w(e_{xy}) = d(x, y)$, so

$$d_{\text{ind}}(x, y) \leq d(x, y).$$

Step 3: d dominates induced distance. For any configuration $\gamma = (e_{x_0x_1}, \ldots, e_{x_{n-1}x_n})$ from x to y, repeated application of the triangle inequality gives

$$d(x, y) \leq \sum_{i=0}^{n-1} d(x_i, x_{i+1}) = w(\gamma).$$

Taking the infimum over all configurations, $d(x, y) \leq d_{\text{ind}}(x, y)$.

Step 4: Equality and isometry. Combining Steps 2 and 3, $d_{\text{ind}} = d$. Since d is a metric, $d(x, y) = 0 \Rightarrow x = y$, so the zero-distance equivalence relation is trivial and $R/\sim \cong X$. The identity map $X \to R/\sim$ is an isometry. □

Remark 7.2 (Expressibility, not canonicity)*.* Theorem 7.1 establishes *expressibility*: every finite metric space can be encoded as a relational system. It does not establish *canonicity*: there is no unique or preferred encoding, and the choice of encoding is itself a contextual decision.

This distinction is important for the paper's central philosophical position. The introduction argues that metrics are external contextual parameters, not intrinsic to geometric objects. The reconstruction theorem is consistent with this position, but the consistency requires care. In the proof, the metric $d(x, y)$ is assigned as an edge weight $w(e_{xy})$ — the metric appears as a primitive input to the relational system. This does not mean the metric is intrinsic to the geometric object; it means that a specific *choice* of external metric has been encoded into a specific relational presentation. A different choice of metric would yield a different relational presentation of the same underlying point set.

The correct interpretation is therefore: any metric can be expressed relationally, but the relational expression always reflects a contextual choice. The framework does not make the metric intrinsic — it makes explicit the choice that classical geometry leaves implicit. Different relational presentations of the same point set, with different weight functions, correspond to different structural contexts imposing different metrics through ambient comparison (Remark 6.6).

A skeptic may ask: if the weight values must be supplied as input, is the framework truly deriving the metric, or merely representing a metric that was already given? This question applies equally to classical geometry. The Euclidean metric on $\mathbb{R}^n$ is not derived from the structure of $\mathbb{R}^n$ — it is selected because it matches the physical context being modeled. The framework does not make this selection problem worse; it makes it visible as a selection rather than concealing it as a primitive. The origin of weight values lies in the physical or computational context the system is modeling, not in the abstract relational structure itself. That is not a limitation of the framework — it is the correct description of where measurement comes from in any foundational account of geometry.

This is also why Theorem 7.5 matters: among all relational presentations of a given metric, the minimal ones are those in which no relation is redundant given the others. The minimal presentation is the one that makes the contextual choice as lean as possible — retaining only the relations that the chosen metric genuinely requires and cannot derive from composition of shorter paths.

7.2 Minimal Presentations

The construction in Theorem 7.1 uses a complete directed graph, which is highly redundant. Any metric space admits a minimal relational presentation.

Definition 7.3 (Generating relational system)**.** A relational system $(R, \circ, w)$ with underlying vertex set X *generates* a metric d on X if the induced metric d_{ind} coincides with d.

Definition 7.4 (Redundant relation)**.** A relation $e_{xy} \in R$ is *redundant* if there exists a configuration $\gamma \in \text{Conf}(x, y)$ not using e_{xy} such that $w(\gamma) = d(x, y)$.

Theorem 7.5 (Minimal presentation)**.** *Every finite metric space (X, d) admits a generating relational system $(R, \circ, w)$ with no redundant relations, minimal with respect to edge inclusion among all generating systems.*

Proof. Start from the complete graph construction of Theorem 7.1. Iteratively remove redundant relations until none remain. Since X is finite this process terminates. Removing a redundant relation preserves the induced metric (an alternative configuration of equal length exists). No remaining relation can be removed without increasing some distance, giving minimality. □

Remark 7.6 (Failure modes and edge cases)*.* We document the cases where the framework's machinery requires care.

Pseudometric rather than metric. If the weight function assigns zero weight to a non-identity relation $r \not\sim \text{id}$, the metric emergence theorem produces a pseudometric rather than a metric (Remark 6.5). This is not a failure — it is the correct output for a non-separated relational system. The separation condition (A5) is an explicit hypothesis, not a hidden assumption.

Non-unique minimal presentations. Minimal presentations are not unique whenever the triangle inequality is slack at some triple. If $d(x, z) < d(x, y) + d(y, z)$ for all intermediate points y, then e_{xz} is non-redundant and must appear in every minimal presentation. If $d(x, z) = d(x, y) + d(y, z)$ for some y, then e_{xz} is redundant and may be omitted. Both choices are valid, yielding different minimal presentations of the same metric. The worked example of Section 8.2 illustrates the case where all triangle inequalities are tight, producing a unique minimal path presentation.

Pathological finite metric spaces. Finite metric spaces with no tight triangle inequalities (strictly ultrametric spaces where $d(x, z) \leq \max\{d(x, y), d(y, z)\}$ strictly) have a unique minimal presentation consisting of all $|X|(|X| - 1)$ directed edges, since no multi-step path is ever as short as the direct edge. These are the maximally complex cases: their minimal presentations are complete directed graphs, and no compression is possible.

Remark 7.7 (Non-uniqueness of minimal presentations)*.* Minimal presentations are not unique in general. A simple example illustrates this: the metric space on three points $\{a, b, c\}$ with $d(a, b) = 1$, $d(b, c) = 1$, $d(a, c) = 2$ (a metric line) has two minimal presentations: one with edges e_{ab} and e_{bc} (the metric line traversed left to right), and one where the roles of b-to-a and b-to-c are reversed. Both produce the same induced metric.

In general, the structure of minimal presentations reflects intrinsic metric properties: the presence or absence of direct geodesic relations, and the tightness of the triangle inequality at each triple of points. Understanding the full relationship between distinct minimal presentations of the same metric space — in particular, characterizing when two minimal presentations are related by a natural transformation in $\mathbf{RelSys}_{\sim}$ — is an open problem. It is closely connected to the congruence classification problem of the future directions.

7.3 Scope and Domain of Validity

The theorems of this section establish the framework's reach for finite metric spaces precisely. We state explicitly what is captured, what is approximated, and what lies outside the framework.

Theorem 7.8 (Domain of validity)**.** *The process-relational framework captures the following classes of metric spaces:*

(i) ***Finite metric spaces (exact).*** *By Theorem 7.1 and the categorical equivalence of Theorem 11.10, every finite metric space arises from a relational system, and the correspondence is an equivalence of categories. No approximation is involved.*

(ii) ***Compact metric spaces (GH-approximate).*** *By Theorem 13.2, any compact metric space is the Gromov–Hausdorff limit of a sequence of finite relational systems. The classical space is recoverable as a limit but does not appear as a finite relational object.*

(iii) ***Locally finite non-compact spaces (pointed GH-approximate).*** *By Theorem 13.6, locally finite spaces such as the k-regular tree arise as pointed Gromov–Hausdorff limits of finite relational systems satisfying uniform local finiteness and rooted local convergence.*

The following lie outside the framework's current scope:

(iv) ***Non-locally-finite infinite spaces.*** *Spaces without a uniform bound on the number of neighbours within any fixed radius do not satisfy the hypotheses of Theorem 13.6 and are not captured by the pointed construction.*

(v) ***Smooth Riemannian manifolds.*** *While compact Riemannian manifolds are GH-approximable by finite relational systems, their smooth structure — tangent spaces, differential forms, smooth maps — does not arise within the framework. Whether smooth structure can be encoded relationally is an open question.*

(vi) ***Continuous symmetry groups.*** *Lie groups,* $\mathrm{SO}(n)$*, and the Poincaré group do not appear as objects of the theory; discrete cyclic approximations are available at any resolution but the continuous groups themselves require the real number substrate.*

Proof. Items (i)–(iii) are proved by the theorems cited. Items (iv)–(vi) follow from the framework's foundational commitment to finite discrete structure: they are classes that require either infinite vertex sets without local finiteness bounds, or the real-number substrate, neither of which is primitive in the framework. □

Remark 7.9 (Boundary cases). The boundary between cases (i)–(iii) and cases (iv)–(vi) is precise: a metric space is exactly capturable if and only if it is finite; it is GH-approximable if and only if it is compact or locally finite. The remaining cases require structure that exceeds what finite relational systems can provide. These are not vague incompleteness but identifiable limits. In each case the limitation points to a concrete research direction: extending the framework to length spaces, formalizing smooth structure relationally, or characterizing which Lie group actions admit discrete relational approximations.

8 Examples

8.1 The Discrete Circle

Let $R = \{a\}$ with total composition $a \circ a^k = a^{k+1}$. Configurations are:

$$\mathrm{id},\ a,\ a^2,\ \ldots,\ a^k,\ \ldots$$

Fix $n \geq 1$ and define the congruence generated by the single law:

$$a^n \sim \mathrm{id}.$$

Verification. This is a congruence: if $a^k \sim a^{k'}$ and $a^\ell \sim a^{\ell'}$, then $k \equiv k' \pmod{n}$ and $\ell \equiv \ell' \pmod{n}$, so $k + \ell \equiv k' + \ell' \pmod{n}$, giving $a^k \circ a^\ell \sim a^{k'} \circ a^{\ell'}$.

Quotient. By Theorem 5.2:

$$\mathcal{C}/{\sim} = \{[a^0], [a^1], \ldots, [a^{n-1}]\} \ \cong\ \mathbb{Z}/n\mathbb{Z},$$

with $[a^k] \circ [a^\ell] = [a^{k+\ell \bmod n}]$.

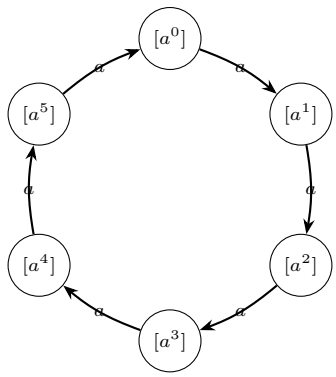


Figure 1: The discrete circle $\mathbb{Z}/6\mathbb{Z}$ as a relational quotient. Each node is an object (equivalence class); each arrow is the generator a. The relational law $a^6 \sim \mathrm{id}$ forces the structure to close into a cycle. No spatial embedding is assumed.

Geometric interpretation. This cyclic structure is the *discrete circle of resolution* n. It encodes the defining topological feature of the circle—closure under a uniform, repeating process—purely through the relational law $a^n \sim \mathrm{id}$. The n objects $[a^0], \ldots, [a^{n-1}]$ correspond to n equally distributed positions on the circle; the composition law encodes the fact that stepping around the circle returns to the starting point. No embedding in a plane, no arc length, and no angular coordinate is used.

Remark 8.1 (Resolution is contextual, not intrinsic). The integer n is not an intrinsic property of the circle. It is a parameter expressing the *resolution* of the relational system relative to a chosen reference step. Two systems with different values of n represent the same relational shape observed at different comparative granularities. The circle as a relational object is the *pattern*—closure under a uniform process—not a fixed number of steps. For example, applying a reference step of one centimetre to a circle of circumference one metre yields $n = 100$; applying a reference step of one millimetre yields $n = 1000$. The relational object — the closure pattern — is unchanged; only the resolution of the external comparison differs.

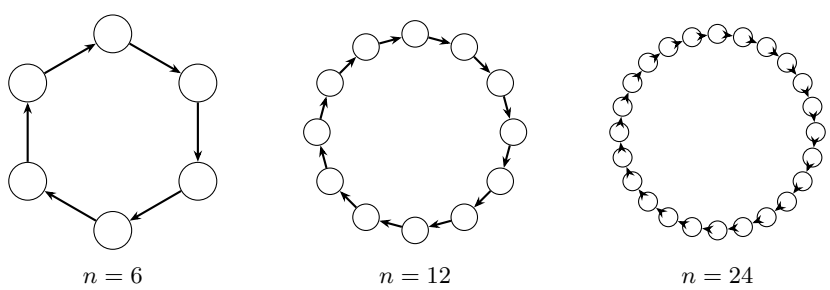


Figure 2: The same relational circle at three resolutions: $n = 6$, $n = 12$, and $n = 24$. Each is the quotient $\mathbb{Z}/n\mathbb{Z}$ arising from the single law $a^n \sim \mathrm{id}$. The relational identity—closure under a uniform process—is identical across all three; only the resolution parameter n differs. Increasing n produces a finer discrete structure that more closely resembles a smooth circle at any given scale of observation, but no limit is taken: each is a complete, exact geometric object in its own right.

Remark 8.2 (Compositional pattern determines geometric identity). The geometric identity of an object $[C] \in \mathcal{C}/\!\sim$ is determined entirely by the congruence $\sim$—that is, by the compositional pattern of the relational laws. External parameters such as resolution n, scale, and positional reference contribute nothing to this identity; they belong to the external context of comparison, not to the object.

A direct consequence: in the discrete circle construction, the relational law $a^n \sim \mathrm{id}$ and the cyclic structure $\mathbb{Z}/n\mathbb{Z}$ are equivalent descriptions of the same geometric entity. Knowing the compositional law

determines the object; knowing the object up to relational indistinguishability determines the law. The map from congruence classes of relational laws to isomorphism classes of quotient objects is a bijection.

This equivalence has a further consequence for geometric classification. Two relational systems may be *extensionally equivalent*—producing isomorphic quotients—without being *intensionally identical*— having the same laws. Extensional equivalence is the geometric notion: two systems are the same shape if no relational test distinguishes them. Intensional identity is a property of the description. The gap between the two is precisely where the classification of geometric structures becomes non-trivial.

Example 8.3 (Distance on the discrete circle)**.** We apply Theorem 6.3 to the discrete circle $\mathbb{Z}/n\mathbb{Z}$. Assign unit weight to the generator: $w(a) = 1$, so $w(a^k) = k$.

The distance between objects $[a^j]$ and $[a^k]$ is the minimum-weight path between them. Two directed paths connect any two objects on the cycle: the *forward path* of length $|k-j| \bmod n$ steps, and the *backward path* of length $n - (|k-j| \bmod n)$ steps. The metric selects the shorter:

$$d\big([a^j],[a^k]\big) \;=\; \min\big(\,|k-j| \bmod n,\;\; n-(|k-j| \bmod n)\,\big).$$

For $n = 6$: the distance from $[a^0]$ to $[a^2]$ is $\min(2,4) = 2$; the distance from $[a^0]$ to $[a^5]$ is $\min(5,1) = 1$, since the backward step is shorter. The metric reflects the circular topology: the maximum distance between any two objects is $\lfloor n/2 \rfloor$.

No coordinates, no arc length, and no embedding in a plane are used. The metric is derived entirely from the relational weight function and finite path optimization, as guaranteed by Theorem 6.3.

Corollary 8.4 (Circle as well-formed closed object)**.** *The discrete circle $\mathbb{Z}/n\mathbb{Z}$ is a well-formed closed object in the sense of Proposition 5.4: the congruence $a^n \sim \mathrm{id}$ satisfies closure, and the n classes $[a^0], \ldots, [a^{n-1}]$ are pairwise inequivalent (uniqueness). Open shapes such as the line arise when no periodicity law is imposed; see Proposition 5.6.*

8.2 End-to-End Worked Example: A Four-Point Metric Space

We demonstrate the full pipeline of the framework on a concrete finite metric space: presentation, reduction to minimal form, metric reconstruction, and comparison with the classical description. This example is chosen to be non-trivial — the minimal presentation is not the complete graph — while remaining fully explicit.

The metric space. Let $X = \{p, q, r, s\}$ be four points with distances:

$$d(p,q) = 1, \quad d(q,r) = 1, \quad d(r,s) = 1, \quad d(p,s) = 3,$$

$$d(p,r) = 2, \quad d(q,s) = 2.$$

This is the metric on four equally-spaced collinear points — a metric line segment of length 3 with three unit steps.

Step 1: Complete graph presentation. By Theorem 7.1, the canonical presentation uses a directed complete graph on $\{p, q, r, s\}$ with twelve edges e_{xy} for every ordered pair (x, y), weighted by $d(x, y)$. This generates the correct metric but is maximally redundant.

Step 2: Identifying redundant relations. A relation e_{xy} is redundant if there exists an alternative configuration of the same weight connecting x to y.

Check e_{pr}: the path $e_{pq} \circ e_{qr}$ has weight $1 + 1 = 2 = d(p, r)$. So e_{pr} is redundant.

Check e_{qs}: the path $e_{qr} \circ e_{rs}$ has weight $1 + 1 = 2 = d(q, s)$. So e_{qs} is redundant.

Check e_{ps}: the path $e_{pq} \circ e_{qr} \circ e_{rs}$ has weight $1 + 1 + 1 = 3 = d(p, s)$. So e_{ps} is redundant.

By symmetry ($w(e_{xy}) = w(e_{yx})$), the reverse edges e_{rp}, e_{sq}, e_{sp} are redundant by the same argument. After removing all six redundant edges, the minimal presentation contains exactly six edges: $e_{pq}, e_{qp}, e_{qr}, e_{rq}, e_{rs}, e_{sr}$.

Step 3: Verifying the minimal presentation. No edge in $\{e_{pq}, e_{qp}, e_{qr}, e_{rq}, e_{rs}, e_{sr}\}$ is redundant. Each connects adjacent points with no shorter alternative path between them (removing e_{pq}, for instance, would make p unreachable from q by a path of weight 1, since the only remaining path from p to q uses e_{qp}^{-1} — the same edge). The minimal presentation is a path graph with unit-weight edges in both directions.

Step 4: Metric reconstruction. The induced metric from the minimal presentation is:

$$d_{\text{ind}}(x, y) = \min_{\gamma \in \text{Conf}(x,y)} w(\gamma),$$

computed by shortest-path over the six-edge graph. This gives the same values as the original metric: $d_{\text{ind}}(p, r) = 2$, $d_{\text{ind}}(p, s) = 3$, etc. Reconstruction is exact.

Step 5: Classical comparison. Classically, four collinear points are described by specifying coordinates — say $p = 0$, $q = 1$, $r = 2$, $s = 3$ on the real line — and deriving distances as absolute differences. The relational description is coordinate-free: no ambient line, no embedding, no reference origin. The metric emerges entirely from the composition structure of six primitive relations. The classical description requires an ambient $\mathbb{R}$; the relational description requires four objects and six rules.

Observation on tightness. The triangle inequality is tight at every triple in this example: $d(p, r) = d(p, q) + d(q, r)$, $d(q, s) = d(q, r) + d(r, s)$, and $d(p, s) = d(p, q) + d(q, r) + d(r, s)$. Tightness at every triple is exactly the condition under which all multi-step edges are redundant — so tight triangle inequalities correspond precisely to sparse minimal presentations. This illustrates the geometric meaning of minimality: the presentation is minimal when the metric is fully explained by the direct relations, with no shortcut distances needed.

8.3 The Discrete Torus

Let $R = \{a, b\}$. Impose the congruence generated by:

$$a \circ b \sim b \circ a, \quad a^n \sim \text{id}, \quad b^m \sim \text{id}.$$

Geometric interpretation. The commutativity law $a \circ b \sim b \circ a$ encodes two *independent* relational directions that do not interact. The periodicity laws $a^n \sim \text{id}$ and $b^m \sim \text{id}$ close each direction into a cycle. The combination produces a two-dimensional discrete surface that wraps in both directions simultaneously: a torus. No ambient $\mathbb{R}^3$, no embedding, and no notion of "surface" in the classical sense is used; the topology arises entirely from the relational laws.

By Theorem 5.2:

$$\mathcal{C}/\!\sim \;\cong\; \mathbb{Z}/n\mathbb{Z} \;\times\; \mathbb{Z}/m\mathbb{Z}.$$

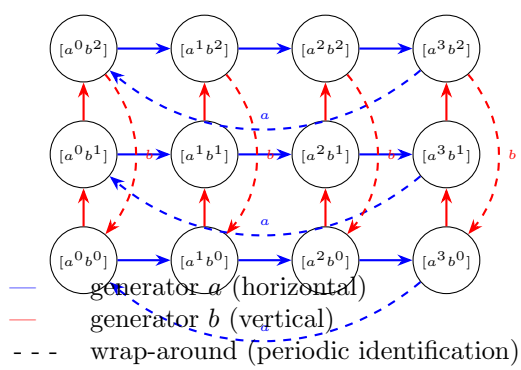


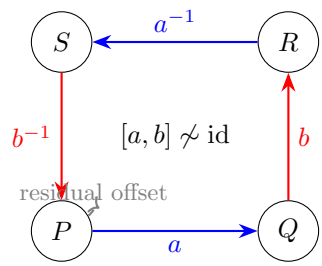
Figure 3: The discrete torus $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ as a relational quotient. Dashed arrows indicate the periodic identifications $a^4 \sim \text{id}$ (horizontal) and $b^3 \sim \text{id}$ (vertical). Commutativity $ab \sim ba$ means the order of traversal is irrelevant. No embedding space is assumed.

8.4 Relational Curvature via Noncommutativity

Let $R = \{a, b, a^{-1}, b^{-1}\}$ with $a \circ a^{-1} \sim \text{id}$ and $b \circ b^{-1} \sim \text{id}$. Suppose $a \circ b \not\sim b \circ a$, and define the *relational commutator*:

$$[a, b] \;=\; a \circ b \circ a^{-1} \circ b^{-1}.$$

Geometric interpretation. The commutator measures the discrepancy between two paths with the same start and end: move first by a then b, versus first by b then a. If $[a, b] \sim \mathrm{id}$, the two paths are indistinguishable and the relational structure is locally flat. If $[a, b] \not\sim \mathrm{id}$, the two paths lead to observationally distinct outcomes: traversing the loop $a \circ b \circ a^{-1} \circ b^{-1}$ does not return to the starting state. This failure of closure is the discrete relational analogue of *holonomy*—the accumulated effect of parallel transport around a closed loop—which in classical differential geometry is the signature of nonzero curvature.



Non-trivial commutator: the loop does not close.

Figure 4: Relational curvature as noncommutativity. The loop $a \circ b \circ a^{-1} \circ b^{-1}$ travels from P to Q to R to S and back. If $[a, b] \not\sim \mathrm{id}$, the final state is observationally distinct from the initial state: the loop has a nontrivial holonomy, encoding discrete curvature. When $[a, b] \sim \mathrm{id}$, the structure is locally flat (discrete torus case).

Curvature in this framework is a discrete, finitely computable property: check whether the commutator $[a, b]$ is equivalent to id under the congruence. No differential calculus, no tangent spaces, and no smoothness assumptions are required.

Remark 8.5 (Relation to noncommutative geometry). The use of noncommutativity to encode geometric curvature connects to the program of noncommutative geometry initiated by Connes, where the commutativity of coordinate algebras is relaxed to capture curved and quantum spaces. In the present framework, the same idea appears in a discrete, purely relational setting: the commutator $[a, b]$ is a measurable relational observable, not an algebraic construction over a function algebra. The framework thus makes the noncommutative geometry intuition fully discrete and finite.

Remark 8.6 (Curvature as a special case of pseudo-periodicity). The commutator $[a, b]$ is an instance of the pseudo-periodicity of Proposition 5.6: traversing the loop $a \circ b \circ a^{-1} \circ b^{-1}$ does not return to id but to the shifted state $[a, b]$. This is case (ii) of the taxonomy, with $n = 4$ (the loop has four steps) and $\phi = [a, b]$. Curvature is thus a two-generator instance of the same algebraic structure that produces helices and spirals in the one-generator case: a composition that shifts rather than closes.

Remark 8.7 (The discrete-to-continuous curvature gap). The commutator $[a, b]$ detects the presence and direction of curvature — whether a loop closes, and what the residual shift is. This is a correct and computable discrete analogue of holonomy.

A harder question, not resolved by the present framework, is whether and how this discrete curvature measure converges to the classical curvature tensors of Riemannian geometry — Gaussian curvature, Ricci curvature, sectional curvature — under refinement of the relational system.

The Gromov–Hausdorff bridge theorems of Section 13 establish metric convergence: sequences of relational systems can approximate compact metric spaces arbitrarily well in the GH sense. But curvature is a second-order quantity. Metric convergence does not automatically imply curvature convergence, and the conditions under which a sequence of commutator-based discrete curvature measures converges to a smooth curvature tensor are not established here.

This gap is a concrete target for future work. The natural setting for the convergence question is the theory of Alexandrov spaces and curvature-dimension conditions, where curvature bounds are defined directly in terms of triangle comparison rather than differential operators. Whether the commutator-based curvature can be connected to these synthetic curvature conditions, and whether that connection is preserved under pointed GH limits, is an open problem that would substantially extend the framework's reach into differential geometry.

Proposition 8.8 (Triangle defect and δ-hyperbolicity)**.** *Let $(\mathcal{O}, d)$ be the metric space induced by a relational system with inverses. Define the* triangle defect*:*

$$\kappa(x, y, z) := d(x, y) + d(y, z) - d(x, z) \quad \text{for } x, y, z \in \mathcal{O}.$$

The following are equivalent up to constants:

(i) $\kappa(x, y, z) \leq \delta$ *for all* $x, y, z \in \mathcal{O}$*;*

(ii) $(\mathcal{O}, d)$ *is* δ'*-hyperbolic in the sense of Gromov, with* $\delta' \asymp \delta$*.*

Proof. $(i) \Rightarrow (ii)$: Assume $\kappa(x, y, z) \leq \delta$ for all triples, i.e. $d(x, z) \geq d(x, y) + d(y, z) - \delta$. Take any geodesic triangle with vertices x, y, z and geodesic sides, and pick a point p on the side $[x, y]$ with $d(x, p) = a$, $d(p, y) = b$, $a + b = d(x, y)$. Applying the defect bound to the triple (x, p, z) gives $d(p, z) \leq d(x, z) - a + \delta$. Comparing with the path through y gives $d(p, z) \leq b + d(y, z)$. Combining these two bounds via the standard thin-triangle argument shows that p lies within $O(\delta)$ of $[x, z] \cup [y, z]$. Hence triangles are $O(\delta)$-thin and $(\mathcal{O}, d)$ is $O(\delta)$-hyperbolic.

$(ii) \Rightarrow (i)$: Assume $(\mathcal{O}, d)$ is δ'-hyperbolic. Apply the four-point condition with base point $o = y$:

$$d(x, z) + d(y, y) \leq \max\{d(x, y) + d(z, y),\ d(x, y) + d(y, z)\} + \delta'.$$

Since $d(y, y) = 0$ and the maximum equals $d(x, y) + d(y, z)$, this gives $d(x, z) \leq d(x, y) + d(y, z) + \delta'$, i.e. $\kappa(x, y, z) \leq \delta'$. □

Remark 8.9 (Curvature types via triangle defect)*.* The triangle defect κ classifies the metric geometry of a relational system by the behavior of geodesic triangles. When $\kappa = 0$ identically, the triangle inequality is always tight, which characterizes tree-like (0-hyperbolic) structures: geodesics compose without residual, and the metric is an ultra-metric in the limit. When $\kappa \leq \delta$ uniformly for some $\delta > 0$, the space is δ-hyperbolic — its large-scale geometry is negatively curved in the sense of Gromov, connecting directly to geometric group theory and the geometry of the regular tree examples of Section 8. When κ is large or unbounded, the space contains positive-curvature behavior; the framework detects this through the presence of short non-trivial loops, as in the icosahedral sphere example.

8.5 Pseudo-Periodicity and the Taxonomy of Closure

The examples so far — the circle, the torus, the open line — represent two extremes of the closure spectrum: fully closed (periodicity law present) and fully open (no periodicity law). A natural intermediate category exists between them.

Definition 8.10 (Pseudo-Periodic Relational System)**.** A relational system $(R, \circ)$ with generator a and congruence $\sim$ is *pseudo-periodic* with period n and *shift* $\phi \in \mathcal{C}/\sim$ if:

$$a^n \sim \phi, \quad \phi \not\sim \mathrm{id}.$$

Each full traversal of n steps does not return to the identity but to a consistently shifted relational state ϕ. The next traversal begins from ϕ, advances n steps, and arrives at $\phi \circ \phi = \phi^2$. Successive traversals accumulate the shift.

The shift ϕ is a fixed relational transformation applied at the end of each cycle. Its algebraic character determines the geometry of the resulting object:

- If ϕ encodes a *translation* along an independent axis, the shape advances forward with each cycle. Combined with a closed oscillation, this produces a *wave*: periodic in one direction, open in another.
- If ϕ encodes a *scaling*, each cycle is larger than the previous by a constant factor. This produces a *logarithmic spiral*: the same angular pattern repeated at increasing scale.
- If ϕ encodes a *rotation* about a propagation axis, each cycle is rotated relative to the previous. This produces a *helix*: constant angular advance per cycle along the rotation axis.

Example 8.11 (Wave as product of closed and open systems)**.** Let $R = \{a, b, b^{-1}\}$ with $b \circ b^{-1} \sim \mathrm{id}$. Impose the congruence generated by:

$$b^m \sim \mathrm{id}, \quad \text{no periodicity law on } a.$$

The b-direction is closed with period m; the a-direction is open. The quotient is:

$$\mathcal{C}/\!\sim \;\cong\; \mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}.$$

This is an infinite sequence in the a-direction — the propagation axis — combined with a finite cycle in the b-direction — the oscillation axis. The result is a discrete wave: periodic in amplitude, unbounded in propagation. The apparent infinity of the wave is entirely in the a-direction and is, as with the line, the absence of a closure law rather than the presence of an infinite object.

Example 8.12 (Helix as pseudo-periodic system)**.** Let $R = \{a, b\}$. Impose:

$$a^n \sim b, \quad b^k \sim \mathrm{id}.$$

The first law makes the system pseudo-periodic in a with shift b. The second closes the shift itself after k accumulations. Each full traversal of n steps in a advances one step in b; after k such traversals, the b-direction closes. The quotient encodes a discrete helix: n steps per turn, k turns before closure.

Remark 8.13 (Pseudo-periodicity in the taxonomy)*.* Pseudo-periodic systems constitute the intermediate case of Proposition 5.6: neither fully closed ($a^n \sim \mathrm{id}$) nor fully open (no periodicity law), but shifting ($a^n \sim \phi$, $\phi \not\sim \mathrm{id}$). The three cases of the taxonomy are mutually exclusive and exhaustive for single-generator systems.

Remark 8.14 (Connection to semidirect products)*.* Pseudo-periodic relational systems correspond algebraically to *semidirect products*. If the shift ϕ acts on the cyclic structure by an automorphism φ, the resulting quotient is the semidirect product $\mathbb{Z}/n\mathbb{Z} \rtimes_\varphi \langle\phi\rangle$. The helix, the screw motion, and the glide reflection are all semidirect products of this form. The relational framework recovers these classical symmetry structures as special cases of pseudo-periodicity, without requiring the ambient space in which they are classically defined.

Remark 8.15 (Continuous symmetries are absent by design)*.* The symmetry structures arising in this framework are discrete: cyclic groups, finite semidirect products, and their combinations. Continuous symmetry groups — Lie groups, the rotation group $\mathrm{SO}(n)$, the Poincaré group — do not appear, because continuous structures are absent from the framework by design.

This is a deliberate foundational choice, not an omission. Continuous symmetries require the real number substrate and the associated machinery of Lie theory. In the present framework, what classical geometry describes as a continuous rotation is represented as a discrete cyclic action at a specified resolution. The group $\mathrm{SO}(2)$, for instance, is not an object in the framework; the cyclic group $\mathbb{Z}/n\mathbb{Z}$ at resolution n is. As n increases, the discrete group approximates the continuous one more finely, but the limit is never taken.

Whether this restriction is physically appropriate depends on the domain of application. For quantum gravity and discrete spacetime models, discrete symmetry groups are natural. For classical mechanics and field theory, where continuous symmetries are central, the framework provides a discrete approximation that may be refined to any required precision but does not reproduce the continuum exactly. This is a known and accepted limitation of any strictly finite framework.

Proposition 8.16 (Symmetry pattern from the generating function)**.** *Let $(R, \circ)$ be a pseudo-periodic relational system with generator a, period n, and shift ϕ. The complete symmetry pattern of the object $\mathcal{C}/\!\sim$ is determined by (n, ϕ) alone, without constructing or inspecting the object. Specifically:*

(i) ***Orbit structure.*** *The shift ϕ acts on the n positions $[a^0], \ldots, [a^{n-1}]$ of each cycle as a permutation σ_ϕ. Object $[a^j]$ in cycle c maps to $[\phi^c \circ a^j]$ in cycle $c+1$. The orbit of any position under successive cycles is directly computable from σ_ϕ.*

(ii) ***Higher-level periodicity.*** *If ϕ has finite order k — that is, $\phi^k \sim \mathrm{id}$ — then the full object repeats after k complete cycles, giving a higher-level period of kn steps. If ϕ has infinite order, the object has no global periodicity but retains local symmetry within each cycle of n steps.*

(iii) ***Symmetry group.*** *The symmetry group of the object is:*

$$G \cong \mathbb{Z}/n\mathbb{Z} \rtimes_{\sigma_\phi} \langle\phi\rangle,$$

where $\langle\phi\rangle \cong \mathbb{Z}/k\mathbb{Z}$ if ϕ has finite order k, and $\langle\phi\rangle \cong \mathbb{Z}$ otherwise. This group is fully determined by n, the order of ϕ, and the permutation σ_ϕ.

Proof. **(i).** By definition of pseudo-periodicity, $a^n \sim \phi$. The action of ϕ on $[a^j]$ is given by left composition: $\phi \circ [a^j] = [\phi \circ a^j]$. Since ϕ is a fixed element of $\mathcal{C}/\sim$, this action is a well-defined endomorphism of the n-cycle. That it is a permutation follows from the invertibility of ϕ in an invertible relational system.

(ii). If $\phi^k \sim \mathrm{id}$, then after k cycles the accumulated shift returns to the identity: $[a^{kn}] \sim [\phi^k \circ a^0] \sim [a^0]$. The object closes at step kn. If no such k exists, all cycles produce distinct relational states and no global closure occurs.

(iii). The symmetry group is generated by a (the within-cycle step) and ϕ (the between-cycle shift), subject to $a^n \sim \phi$ and the order of ϕ. The semidirect product structure follows from the action of ϕ on the n-cycle by σ_ϕ. □

Corollary 8.17 (Compact encoding of pseudo-periodic objects)**.** *A pseudo-periodic relational object with period n, shift ϕ of order k, and permutation action σ_ϕ is completely encoded by the triple (n, k, σ_ϕ). The full object — including its symmetry group, orbit structure, and higher-level periodicity — is recoverable from this triple by the construction of Proposition 8.16.*

In particular, the encoding cost is independent of the resolution parameter: the same triple (n, k, σ_ϕ) encodes the object at any resolution, with n supplying only the within-cycle step count. The intrinsic geometric information is contained in k and σ_ϕ; the resolution n is, as established in Remark 8.1, an external contextual parameter.

Remark 8.18 (Symmetry without construction). Proposition 8.16 and Corollary 8.17 establish a property that is non-trivial from a classical perspective: the complete symmetry pattern of a geometric object is readable directly from its generating laws, without constructing the object, without embedding it in a space, and without performing geometric analysis. The symmetry is algebraic and the algebra is the object.

This is a direct consequence of the description/object equivalence of Remark 8.2: because the congruence and the object are equivalent descriptions of the same thing, any property derivable from the congruence is a geometric property of the object. Symmetry is one such property. It is encoded in the laws before the object exists.

Remark 8.19 (Connection to universality and categorical equivalence). The compact encoding of Corollary 8.17 situates pseudo-periodic objects within the broader framework of Sections 7 and 11. The encoding triple (n, k, σ_ϕ) determines a finite relational system whose induced metric is a specific finite metric space; by Theorem 7.1, this metric space is uniquely recoverable from the relational data. Under the categorical equivalence of Theorem 11.10, the symmetry group $G \cong \mathbb{Z}/n\mathbb{Z} \rtimes_{\sigma_\phi} \langle\phi\rangle$ derived from the relational laws corresponds precisely to the automorphism group of the induced metric space in $\mathbf{LawMet}_{\mathrm{fin}}$.

Thus the pseudo-periodicity analysis is not an isolated result but an instance of the general principle that intrinsic geometric properties — including symmetry — are fully determined by the relational presentation and can be read off without constructing the geometric object explicitly.

8.6 Negative Curvature: The Infinite Regular Tree

We construct a relational system whose induced metric exhibits hyperbolic-type geometry: unique geodesics, exponential volume growth, and 0-thin triangles. This is the canonical discrete model of negative curvature.

Fix an integer $k \geq 2$. Define a relational system as follows. The vertex set V consists of all finite words over the alphabet $\{1, \ldots, k\}$, including the empty word $\varnothing$. For each vertex v and symbol $i \in \{1, \ldots, k\}$, include a relation $e_{v,i} : v \to vi$ (appending the symbol), and a reverse relation $\bar{e}_{v,i} : vi \to v$ (removing the symbol). Assign unit weights $w(e_{v,i}) = w(\bar{e}_{v,i}) = 1$.

This defines the infinite k-regular tree rooted at $\varnothing$.

Proposition 8.20. *The induced metric satisfies:*

$$d(v, w) = |v| + |w| - 2|v \wedge w|,$$

where $|v|$ denotes word length and $v \wedge w$ is the longest common prefix of v and w.

Proof. Any configuration from v to w must ascend to the common ancestor $v \wedge w$ and then descend. The shortest path does exactly this, with length $(|v| - |v \wedge w|) + (|w| - |v \wedge w|)$. No shorter path exists since the structure is a tree. □

Theorem 8.21 (Hyperbolic features of the tree)**.** *The relational system (V, d) satisfies:*

(i) ***Unique geodesics.*** *Between any two vertices there is exactly one shortest configuration.*

(ii) ***Exponential growth.*** *The number of vertices at distance n from the root is k^n.*

(iii) **0-*thin triangles.*** *For any three vertices, each side of the geodesic triangle is contained in the union of the other two.*

Proof. (i) The tree has no cycles, so there is a unique simple path between any two vertices. (ii) Each vertex has exactly k children, giving k^n vertices at depth n. (iii) In a tree, every geodesic triangle is degenerate: one side lies on the path connecting the other two. □

Remark 8.22. Exponential volume growth and 0-thin triangles are the defining combinatorial signatures of negative curvature in Gromov's sense. This example shows that hyperbolic-type geometry arises from purely relational data without any ambient curved space: the curvature is a consequence of the branching structure of the relational composition, not of any background geometry.

8.7 Positive Curvature: Triangulated Sphere

We construct a finite relational system exhibiting characteristic features of positive curvature: reconverging geodesics and global distance bounds. We work with a fixed finite triangulation rather than a limit.

Let T be any triangulation of the 2-sphere S^2 with vertex set V_T and edge set E_T. Define a relational system (G_T, w_T): vertices are elements of V_T; relations are edges of E_T (taken as bidirectional); weights are set to $w_T(e) = 1$ for all edges.

Proposition 8.23. *In the relational system (G_T, w_T):*

(i) The induced metric d_T counts the minimum number of triangulation edges in any path between two vertices.

(ii) The diameter of G_T is finite and bounded by the combinatorial structure of the triangulation.

Proof. (i) Each edge has weight 1, so path length counts edges. (iii) Any connected finite graph has finite diameter. □

Example 8.24 (Geodesic non-uniqueness: the icosahedral graph)**.** The icosahedral triangulation has 12 vertices, 30 edges, and 20 triangular faces. Each vertex is connected to 5 neighbours; the icosahedral graph has diameter 3. By the icosahedral symmetry group $A_5 \times \mathbb{Z}/2\mathbb{Z}$, every vertex has a unique antipodal vertex at distance 3.

Let v be any vertex and $\bar{v}$ its antipodal. There are exactly 6 distinct shortest configurations (paths of length 3) from v to $\bar{v}$, corresponding to the 6 pairs of opposite faces adjacent to v. No shorter path exists since the diameter is 3.

This gives 6 distinct minimal configurations connecting the same pair of objects — the relational analogue of multiple geodesics between antipodal points on a positively curved surface.

Remark 8.25. The key distinction from the tree (negative curvature) is the *reconvergence* of paths: on the icosahedral graph, paths that diverge from v eventually reconverge at $\bar{v}$. Curvature in the relational framework is the commutator structure of Section 8.4; positive curvature manifests as the global constraint that long "straight" paths cease to be minimal beyond a certain scale, a property that emerges directly from the relational composition rules of the triangulation.

8.8 Flat Geometry: Integer Grid

We construct a finite relational system exhibiting flat (zero curvature) geometry. Let $n \geq 1$. Define a relational system (G_n, w_n) as follows:

- Vertex set $V_n := \{0, 1, \ldots, n\}^2 \subset \mathbb{Z}^2$ (a finite square grid);
- Edge set: connect each vertex (i, j) to its eight neighbours — four axis-aligned and four diagonal;
- Weight function: $w_n(e) := 1$ for all edges (axis and diagonal alike).

Proposition 8.26. *The induced metric d_n on (G_n, w_n) is the Chebyshev (ℓ^∞) distance:*

$$d_n\big((i_1, j_1), (i_2, j_2)\big) = \max\big\{|i_1 - i_2|,\ |j_1 - j_2|\big\}.$$

Proof. Any path from (i_1, j_1) to (i_2, j_2) must cover a horizontal displacement of $|i_1 - i_2|$ and a vertical displacement of $|j_1 - j_2|$. Each step reduces at least one coordinate discrepancy by 1. The minimum number of steps needed is therefore $\max\{|i_1 - i_2|, |j_1 - j_2|\}$, achieved by using diagonal steps to reduce both coordinates simultaneously. No shorter path exists. □

Remark 8.27 (Flatness as commutative composition). The Chebyshev metric on the integer grid is a flat discrete metric: the two generators a (horizontal) and b (vertical) commute, $a \circ b \sim b \circ a$, which is precisely the condition $[a, b] \sim \mathrm{id}$ of Section 8.4. The torus of Section 8.3 is this flat grid with periodic boundary conditions, confirming that flatness is a property of the relational composition rules, not of any ambient space.

The Chebyshev metric is one of several discrete flat metrics constructible within the framework. Integer grids with axis-only edges yield the ℓ^1 (Manhattan) metric; the Chebyshev metric arises from the full 8-connected grid. Both are exact, rational, and require no approximation of irrational quantities.

9 Categorical Interpretation

The quotient construction has a natural categorical reading that provides conceptual orientation before the formal development of Section 11.

At an informal level:

- Primitive relations in R are *generating morphisms*.
- The set of all configurations forms the *free category* on R: objects are sources and targets of relations, and morphisms are composable chains.
- The congruence $\sim$ *imposes relations* on this free category, quotienting it.
- The quotient $\mathcal{C}/\!\sim$ is the *category presented by generators R and relations $\sim$* — the categorical analogue of a monoid or group presentation.

When every generator has a compositional inverse up to $\sim$, the quotient is a *groupoid*. The circle yields $\mathbb{Z}/n\mathbb{Z}$ as a one-object groupoid; the torus yields $\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$ as a product.

The present framework *contracts* rather than *generalizes* classical category theory: we start with relations alone and derive objects, rather than assuming objects and morphisms as primitives.

The formal development — including Lawvere metric spaces, the metric realization functor, the universal property of the free enriched category, and the equivalence of categories $\mathbf{RelSys}_\sim \simeq \mathbf{LawMet}_{\mathrm{fin}}$ — is given in Section 11.

10 Higher Relational Structure

The framework extends naturally to higher levels.

- *Level 0.* Primitive relations in R are 0-cells.
- *Level 1.* Configurations in $\mathcal{C}$ are 1-morphisms (composable chains of 0-cells).
- *Level 2.* Congruences between configurations are 2-morphisms: identifications between processes.
- *Level k.* Congruences between $(k-1)$-morphisms are k-morphisms, for each $k \geq 2$.

When all morphisms at each level are invertible, this structure is a *strict ω-groupoid*. When finiteness is maintained at each level—that is, the generating data at every level is a finite set— the entire hierarchy remains finite and discrete.

The discrete circle at Level 2 is the fundamental groupoid of a finite directed graph, with configurations as 1-morphisms and identifications between configurations (homotopies in the discrete sense) as 2-morphisms.

Example 10.1 (A 2-morphism on the discrete circle)**.** Consider the discrete circle $\mathbb{Z}/6\mathbb{Z}$ with generator a. Two distinct configurations connect $[a^0]$ to $[a^4]$:

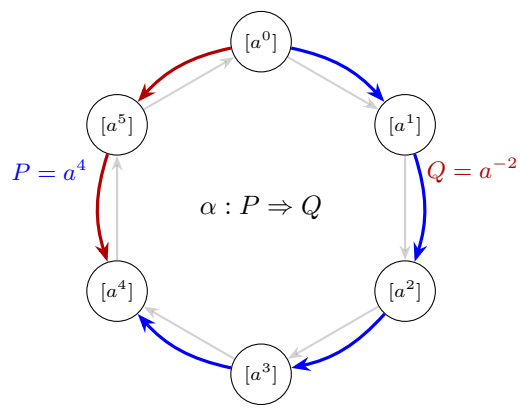
$$P \;=\; a^4 \quad \text{(four forward steps)}, \qquad Q \;=\; a^{-2} \quad \text{(two backward steps)}.$$

Both P and Q are 1-morphisms from $[a^0]$ to $[a^4]$ in the groupoid. They are different processes but produce the same relational outcome.

A 2-*morphism* $\alpha : P \Rightarrow Q$ is a relational identification between these two processes: a witness that P and Q are indistinguishable as paths from $[a^0]$ to $[a^4]$. In the discrete setting this identification is encoded by the congruence:

$$a^4 \;\sim\; a^{-2},$$

which holds because $a^6 \sim \mathrm{id}$ implies $a^4 \circ a^2 \sim \mathrm{id}$, so $a^4 \sim a^{-2}$.



The 2-morphism α is the discrete relational analogue of a homotopy between two paths: it certifies that the two processes are equivalent without constructing a continuous deformation between them. The forward path P (blue, four steps clockwise) and the backward path Q (red, two steps anticlockwise) are algebraically identified by the congruence $a^4 \sim a^{-2}$, which is the 2-morphism. The equivalence is algebraic, computable, and requires no ambient space.

Remark 10.2 (Level structure of the 2-morphism)*.* In the higher categorical framework described above, a 2-morphism should properly be an element of a *second-level* congruence $\sim_2$ defined on the path category, distinct from the first-level congruence $\sim$ that identifies configurations as objects. In the present example, the first-level congruence $a^4 \sim a^{-2}$ serves both roles simultaneously, because the cyclic group $\mathbb{Z}/6\mathbb{Z}$ is simple enough that the two levels collapse: every identification of configurations as objects is also an identification of paths. A full treatment of the two-level structure for more complex relational systems — where $\sim$ and $\sim_2$ are genuinely distinct — is part of the higher categorical development described in the future directions.

11 Enriched Categorical Formulation and Equivalence

The metric realization of a relational system has a natural interpretation in enriched category theory: a relational system is precisely a weighted directed graph, and the induced metric is the hom-structure of the free category enriched over the monoidal poset $([0,\infty], \geq, +, 0)$. This formulation provides the universal characterization of the construction and yields a categorical equivalence with finite metric spaces.

11.1 Lawvere Metric Spaces

Definition 11.1 (Lawvere metric space [1])**.** A *Lawvere metric space* is a set X equipped with a function $d : X \times X \to [0,\infty]$ satisfying, for all $x, y, z \in X$:

$$d(x,x) = 0, \qquad d(x,z) \leq d(x,y) + d(y,z).$$

Symmetry is not required in general. A classical metric space corresponds to a symmetric Lawvere metric space with $d(x,y) = 0 \Rightarrow x = y$.

11.2 Relational Systems as Free Enriched Categories

Definition 11.2 (Free enriched category of a relational system)**.** Given a relational system $(R, \circ, w)$ presented as a weighted directed graph (G, w), define a category enriched over $([0,\infty], \geq, +, 0)$ by:

- Objects: vertices V of G;
- Hom-value: $\mathcal{C}_G(x,y) := \min_{\gamma \in \mathrm{Conf}(x,y)} w(\gamma)$, where the minimum is taken over all configurations from x to y (and equals ∞ if no such configuration exists).

Composition is given by addition: $\mathcal{C}_G(x,z) \leq \mathcal{C}_G(x,y) + \mathcal{C}_G(y,z)$.

Proposition 11.3. *The assignment $(G,w) \mapsto \mathcal{C}_G$ defines a well-formed $([0,\infty], \geq, +, 0)$-enriched category.*

Proof. The triangle inequality for enriched hom-values holds because any concatenation of configurations yields an admissible path: $\mathcal{C}_G(x,z) \leq \mathcal{C}_G(x,y) + \mathcal{C}_G(y,z)$ by taking the concatenation of minimizing paths. The identity condition $\mathcal{C}_G(x,x) = 0$ follows from the empty configuration. □

Theorem 11.4 (Free enriched category: universal property)**.** *The assignment $(G,w) \mapsto \mathcal{C}_G$ is left adjoint to the forgetful functor from $([0,\infty], \geq, +, 0)$-enriched categories to weighted directed graphs.*

Sketch. Given a weighted directed graph (G,w) and an enriched category $\mathcal{C}$, any weight-preserving graph map $\phi : G \to \mathcal{C}$ extends uniquely to an enriched functor $\mathcal{C}_G \to \mathcal{C}$ by sending each configuration to the composite of its image morphisms. This is the universal property of the free construction. □

Remark 11.5. Theorem 11.4 provides the universal characterization requested by the framework's categorical foundations: the construction of $\mathcal{C}_G$ from (G,w) is not ad hoc but is the unique functorial extension satisfying a universal mapping property. Geometry arises as the hom-structure of a universally constructed enriched category.

11.3 Metric Realization Functor

Definition 11.6 (Category of relational systems)**.** Let **RelSys** be the category whose objects are relational systems (G,w) and whose morphisms $F : (G,w) \to (G',w')$ are graph morphisms preserving composition and satisfying $w'(F(e)) \leq w(e)$ for all edges e.

Definition 11.7 (Category of Lawvere metric spaces)**.** Let **LawMet** denote the category of Lawvere metric spaces with 1-Lipschitz maps.

Theorem 11.8 (Metric realization functor)**.** *There is a functor $\mathcal{M}$: **RelSys** $\to$ **LawMet** defined by $\mathcal{M}(G,w) := (V, d_G)$ where $d_G(x,y) = \mathcal{C}_G(x,y)$.*

Proof. For any morphism $F : (G,w) \to (G',w')$ and any configuration γ in G, the image $F(\gamma)$ is a configuration in G' with $w'(F(\gamma)) \leq w(\gamma)$. Taking minima, $d_{G'}(F(x), F(y)) \leq d_G(x,y)$, so $\mathcal{M}(F)$ is 1-Lipschitz. Functoriality with respect to composition and identities is immediate. □

11.4 Equivalence via Localization

We now identify the precise sense in which relational systems and metric spaces carry the same information.

Definition 11.9 (Metric equivalence of relational systems)**.** Two relational systems (G, w) and (G', w') are *metrically equivalent* if there exists a bijection $\phi : V(G) \to V(G')$ with $d_{G'}(\phi(x), \phi(y)) = d_G(x, y)$ for all x, y. Let $\mathbf{RelSys}_\sim$ denote the category obtained from $\mathbf{RelSys}$ by identifying metrically equivalent objects.

Theorem 11.10 (Equivalence of categories)**.** *The metric realization functor induces an equivalence of categories:*

$$\mathbf{RelSys}_\sim \;\simeq\; \mathbf{LawMet}_{\mathrm{fin}},$$

where $\mathbf{LawMet}_{\mathrm{fin}}$ *is the full subcategory of finite metric spaces.*

Proof. **Essential surjectivity.** By Theorem 7.1, every finite metric space (X, d) arises as $\mathcal{M}(G, w)$ for an explicit relational system. Hence $\mathcal{M}$ is essentially surjective onto $\mathbf{LawMet}_{\mathrm{fin}}$.

Full faithfulness. We must show that for any two relational systems (G, w) and (G', w') with induced metrics d_G and $d_{G'}$, the map

$$\mathbf{RelSys}_\sim\big((G, w),\, (G', w')\big) \;\longrightarrow\; \mathbf{LawMet}_{\mathrm{fin}}\big((V, d_G),\, (V', d_{G'})\big)$$

is a bijection.

Injectivity. If two morphisms $F_1, F_2 : (G, w) \to (G', w')$ induce the same 1-Lipschitz map on vertices, then $F_1(v) = F_2(v)$ for all $v \in V$, hence $F_1 = F_2$ as graph morphisms.

Surjectivity. Let $f : (V, d_G) \to (V', d_{G'})$ be a 1-Lipschitz map. We construct a morphism of relational systems realizing f.

By Theorem 7.1, both (G, w) and (G', w') are metrically equivalent (in $\mathbf{RelSys}_\sim$) to their canonical complete-graph presentations: the relational system $(\hat{G}, \hat{w})$ with vertex set V, a direct edge $e_{xy} : x \to y$ for every ordered pair $(x, y) \in V \times V$, and $\hat{w}(e_{xy}) = d_G(x, y)$; and similarly $(\hat{G}', \hat{w}')$ with weights $\hat{w}'(e'_{uv}) = d_{G'}(u, v)$.

Define a graph morphism $\hat{F} : \hat{G} \to \hat{G}'$ by

$$\hat{F}(v) = f(v), \qquad \hat{F}(e_{xy}) = e'_{f(x)f(y)}.$$

This is well-defined since $\hat{G}'$ contains an edge between every ordered pair. We verify the weight condition:

$$\hat{w}'\big(\hat{F}(e_{xy})\big) = \hat{w}'(e'_{f(x)f(y)}) = d_{G'}(f(x), f(y)) \le d_G(x, y) = \hat{w}(e_{xy}),$$

where the inequality holds because f is 1-Lipschitz. Hence $\hat{F}$ is a morphism in $\mathbf{RelSys}$.

Passing to $\mathbf{RelSys}_\sim$, we identify (G, w) with $(\hat{G}, \hat{w})$ and (G', w') with $(\hat{G}', \hat{w}')$, and $\hat{F}$ is a morphism from the first to the second that induces f on metric spaces. This establishes surjectivity.

Therefore $\mathcal{M}$ is an equivalence. □

Remark 11.11 (Relational systems as a presentation theory)*.* Theorem 11.10 is the categorical capstone of the framework. It shows that relational systems and finite metric spaces carry exactly the same information: a relational system is a *presentation* of a metric space, specifying generators and composition rules from which the geometry is reconstructed. Different relational systems presenting the same metric space are related by metric equivalence — a form of gauge freedom in the relational description. The geometry is the invariant content; the relational system is one way of encoding it.

12 Connections to Existing Frameworks

The present framework shares surface similarities with several existing theories, but differs from each in the foundational commitments of Section 1.

What is reformulated and what is genuinely new. A skeptical reader will ask whether the framework discovers new phenomena or repackages known ones. Both occur, and the distinction is worth making explicit.

Reformulated from known theory. The quotient construction (Theorem 5.2) is an instance of the universal property of a category presented by generators and relations — this is standard categorical algebra [6]. The metric emergence theorem (Theorem 6.3) is, in the single-object case, the construction of a word metric on a group presentation; in the multi-object case it is the Cayley graph metric of the quotient groupoid. Both are well-known objects. The categorical equivalence (Theorem 11.10) is a consequence of Lawvere's identification of metric spaces with enriched categories [1] together with the reconstruction theorem.

Genuinely new. The (n, k) classification (Proposition 5.10) — a complete isomorphism invariant for single-generator congruences defined purely in relational language, with intrinsic period and shift order as the invariants — does not appear in the existing literature on presentations of monoids or groups in this form. The pointed GH emergence theorem (Theorem 13.6) with its three explicit conditions (uniform local finiteness, uniform edge bounds, rooted local convergence) is a concrete relational formulation not present in the metric geometry literature. The domain of validity theorem (Theorem 7.8), which draws the boundary between exactly-capturable, GH-approximable, and out-of-scope spaces, is new as an organized statement. And the philosophical framework — the explicit separation of intrinsic relational structure from external contextual parameters (position, size, metric), the expressibility/canonicity distinction, and the identification of infinity as absence of closure rather than primitive object — is an original foundational position, not a restatement of existing work.

The honest summary. The machinery is classical. The organizational framework, the classification results, and the philosophical position are new.

Category theory and enriched categories. The quotient is an instance of a category presented by generators and relations [6, 7]. The key difference is that classical category theory takes objects as primitive data; here, objects are derived. The framework is a *generator-first* approach to categorical structure. Section 11 develops this further via Lawvere's formulation of metric spaces as enriched categories [1], yielding the universal property and categorical equivalence results.

String rewriting systems. The relational system $(R, \sim)$ is formally a string rewriting system, and the quotient is the monoid it presents [10]. The framework extends this by interpreting the presented structure geometrically rather than computationally, and by allowing partial rather than total composition.

Computational implementation. The framework is directly amenable to computational implementation. Relational systems are finite graphs; configurations are directed paths; congruences are equivalence relations on those paths, representable as rewriting rules. Standard algorithms apply without modification: shortest-path algorithms (Dijkstra, Bellman-Ford [3]) compute the metric of Theorem 6.3 exactly; string rewriting normal form algorithms (Knuth-Bendix completion [4]) decide congruence membership and classify quotient structures; graph isomorphism algorithms [5] determine extensional equivalence between relational systems. A concrete computational advantage: geometric equivalence checking — determining whether two relational systems describe the same shape — reduces for single-generator systems to comparing the pair (n, k) by Proposition 5.10, a computation requiring $O(n)$ time rather than the isometry testing required in classical metric geometry. For multi-generator systems, however, determining whether two relational systems are congruence-equivalent is significantly harder: it is related to the graph isomorphism problem [5] and, in the infinite setting, may be undecidable. The classification gap identified in Remark 5.12 thus has a direct computational counterpart — the single-generator case is efficiently decidable, while the general case remains a computationally intensive open problem. The finiteness and discreteness of the framework are not restrictions on expressiveness — they are precisely the properties that make the geometry directly computable in the cases where the classification is understood.

Algebraic topology. The constructions of discrete circles, tori, and commutator-based curvature connect to combinatorial algebraic topology [9, 8]. The classical theory, however, builds on a continuous substrate (simplicial complexes, topological spaces). The present framework achieves analogous constructions without any continuous substrate.

Discrete and combinatorial geometry. Graph-based models of space [11] are structurally compatible with the framework. The framework can be seen as a compositional and algebraic enrichment of graph-theoretic geometry, where edges (relations) carry algebraic structure and equivalence classes (objects) replace vertices as the geometric units.

Directions toward physics. Several approaches to quantum gravity—causal set theory [12], loop quantum gravity, and spin foam models—proceed from the premise that spacetime at the Planck scale is discrete and relational rather than continuous and manifold-like. The present framework is conceptually aligned with this premise and provides a clean algebraic setting in which to formalize relational spacetime structures.

As a concrete illustration, a causal set is a locally finite partially ordered set $(X, \preceq)$ in which $x \preceq y$ means event x causally precedes event y. In the present framework this maps directly: the elements of X are relational configurations, the causal order $\preceq$ is the composition rule, and two causal histories are equivalent if no causal test distinguishes them. The quotient $\mathcal{C}/\sim$ recovers the causal structure as a discrete relational object, with curvature encoded in the commutator structure of causal relations exactly as in Section 8.4. The framework provides the algebraic machinery to reason about causal set geometry — symmetry, holonomy, equivalence — without embedding the causal set in a continuum.

Digital topology and computational geometry similarly operate with finite, discrete data as primitive; the framework offers a compositional foundation for these fields.

Relationship to classical geometry. The present framework does not approximate classical geometry as a limit of finer relational structures. This distinction is important. Taking a limit of discrete relational systems to recover, say, the Euclidean plane would reintroduce the continuum through the back door and would require the same limiting machinery the framework is designed to avoid.

The correct relationship is different. Classical geometries — Euclidean, hyperbolic, spherical — are not limiting cases of relational structures. They are relational structures equipped with specific *external contextual parameters*: a choice of metric, a choice of reference frame, a choice of scale. Remove those choices and the relational substrate remains. Add them back and the classical geometry is recovered, not as a limit but as a parameterization. Different choices of external parameters over the same relational substrate produce different classical geometries. The apparent multiplicity of classical geometries is a consequence of the multiplicity of contextual parameterizations, not a multiplicity of underlying structures.

A finer relational structure — larger n — does produce a discrete object that more closely resembles a classical smooth object at any given scale of comparison. This is a practical observation about approximation quality, not a mathematical limit: the discrete circle $\mathbb{Z}/n\mathbb{Z}$ with $n = 10{,}000$ is observationally indistinguishable from a smooth circle at most scales of physical measurement, but it is not the limit of a sequence converging to the smooth circle. The difference matters foundationally: approximation quality is an external, contextual statement about resolution; the smooth circle as a mathematical object requires actual infinity and does not appear in the framework.

This means the framework is not weaker than classical geometry for lacking limits; it is more general, in the sense that it separates what is intrinsic to a geometric structure from what is supplied by the context of observation.

13 Compatibility with Classical Geometry

The preceding sections establish the relational framework on its own discrete terms. We now address the compatibility question: in what precise sense do classical continuous geometric spaces relate to relational systems? We show that classical spaces arise as *Gromov–Hausdorff limits* of sequences of finite relational systems. This is a bridge result, not a foundational claim: the classical spaces are not the targets the framework was aimed at, but the framework is nonetheless compatible with them in a rigorous sense.

13.1 Metric Approximation by Relational Systems

Definition 13.1 (Metric approximation)**.** Let (X, d) be a compact metric space. A sequence of relational systems (G_n, w_n) with induced metrics (V_n, d_n) and maps $\phi_n : V_n \to X$ *approximates* (X, d) if:

(i) **Density.** $\phi_n(V_n)$ is ε_n-dense in X with $\varepsilon_n \to 0$;

(ii) **Distance approximation.** $|d_n(x,y) - d(\phi_n(x), \phi_n(y))| \le \varepsilon_n$ for all $x, y \in V_n$.

Theorem 13.2 (Gromov–Hausdorff bridge theorem [2])**.** *Let (X,d) be a compact metric space. If (G_n, w_n) approximates (X,d), then*

$$d_{\mathrm{GH}}((V_n, d_n),\, (X,d)) \to 0,$$

where d_{GH} denotes the Gromov–Hausdorff distance.

Proof. We construct an explicit correspondence between (V_n, d_n) and (X,d) and bound its distortion.

Forward map. The given map $\phi_n : V_n \to X$ satisfies $|d_n(x,y) - d(\phi_n(x), \phi_n(y))| \le \varepsilon_n$ for all $x, y \in V_n$.

Reverse map. Since $\phi_n(V_n)$ is ε_n-dense in X, for each $z \in X$ there exists a point $\psi_n(z) \in V_n$ with $d(z, \phi_n(\psi_n(z))) \le \varepsilon_n$. This defines a map $\psi_n : X \to V_n$.

Correspondence. Define the correspondence $\mathcal{R}_n \subseteq V_n \times X$ by

$$\mathcal{R}_n := \big\{(x, \phi_n(x)) \mid x \in V_n\big\} \,\cup\, \big\{(\psi_n(z), z) \mid z \in X\big\}.$$

This is a correspondence: every $x \in V_n$ is related to $\phi_n(x) \in X$, and every $z \in X$ is related to $\psi_n(z) \in V_n$.

Distortion bound. For any pair $(x,z), (x',z') \in \mathcal{R}_n$, we need to bound $d(\phi_n(x), z)$ and $d(\phi_n(x'), z')$. There are two cases depending on which part of the correspondence each pair comes from:

- If (x,z) comes from the first set, then $z = \phi_n(x)$, so $d(\phi_n(x), z) = 0 \le \varepsilon_n$.
- If (x,z) comes from the second set, then $x = \psi_n(z)$, so $d(\phi_n(x), z) = d(\phi_n(\psi_n(z)), z) \le \varepsilon_n$ by the density condition on ψ_n.

In both cases $d(\phi_n(x), z) \le \varepsilon_n$, and likewise $d(\phi_n(x'), z') \le \varepsilon_n$. The triangle inequality then gives:

$$\begin{aligned}
\big|d_n(x,x') - d(z,z')\big| &\le \big|d_n(x,x') - d(\phi_n(x), \phi_n(x'))\big| \\
&\quad + \big|d(\phi_n(x), \phi_n(x')) - d(z,z')\big| \\
&\le \varepsilon_n + d(\phi_n(x), z) + d(\phi_n(x'), z') \\
&\le \varepsilon_n + \varepsilon_n + \varepsilon_n = 3\varepsilon_n.
\end{aligned}$$

Conclusion. The Gromov–Hausdorff distance satisfies $d_{\mathrm{GH}}((V_n, d_n), (X,d)) \le \frac{3}{2}\varepsilon_n$ (by the characterization in terms of correspondences, see [2] Theorem 7.3.25). Since $\varepsilon_n \to 0$, we obtain $d_{\mathrm{GH}}((V_n, d_n), (X,d)) \to 0$. □

Corollary 13.3. *The following classical spaces arise as Gromov–Hausdorff limits of sequences of finite relational systems via Theorem 13.2:*

(i) The circle (S^1, d_{S^1}), via the cyclic graphs $(\mathbb{Z}/n\mathbb{Z}, d_n)$ with d_n the arc-step metric;

(ii) The sphere (S^2, d_{S^2}), via refining geodesic triangulations;

(iii) The Euclidean plane $(\mathbb{R}^2, \|\cdot\|_2)$, locally, via integer grids with diagonal edges.

Tree-like (hyperbolic) spaces are non-compact and require the pointed extension; see Theorem 13.6 and Example 13.7 below.

13.2 Pointed Gromov–Hausdorff Convergence

The compactness assumption in Theorem 13.2 excludes non-compact limits such as the k-regular tree. The pointed Gromov–Hausdorff framework extends compatibility to this case by working with finite balls around a basepoint rather than the full space.

Definition 13.4 (Rooted relational system)**.** A *rooted relational system* is a triple (G, w, v) where (G, w) is a relational system and $v \in V(G)$ is a distinguished basepoint.

Definition 13.5 (Pointed Gromov–Hausdorff convergence)**.** A sequence of rooted relational systems (G_n, w_n, v_n) with induced metrics (V_n, d_n, v_n) *converges in the pointed Gromov–Hausdorff sense* to a pointed metric space (V, d, v) if, for every $R > 0$, the metric balls $B_{d_n}(v_n, R)$ converge in the Gromov–Hausdorff sense to $B_d(v, R)$.

Theorem 13.6 (Pointed GH emergence)**.** *Let (G_n, w_n, v_n) be a sequence of rooted relational systems. Assume:*

(i) ***Uniform local finiteness.*** *There exists $D \in \mathbb{N}$ such that every vertex in every G_n has degree at most D;*

(ii) ***Uniform edge bounds.*** *There exist $0 < a \leq b < \infty$ such that $a \leq w_n(e) \leq b$ for all n and all edges e;*

(iii) ***Rooted local convergence.*** *For every $R > 0$, the balls $B_{d_n}(v_n, R)$ stabilize up to isomorphism as weighted graphs for all sufficiently large n.*

Then there exists a pointed metric space (V, d, v) such that (V_n, d_n, v_n) converges to (V, d, v) in the pointed Gromov–Hausdorff sense.

Sketch. By uniform local finiteness and uniform edge bounds, the metric balls $B_{d_n}(v_n, R)$ are uniformly controlled in size and geometry for each fixed R. By rooted local convergence, they stabilize as weighted graphs for large n. The stabilized balls are consistent across radii: since $B_{d_n}(v_n, R)$ is the restriction of $B_{d_n}(v_n, R+1)$ to vertices at distance at most R from v_n, once both balls have stabilized the restriction of the radius-$(R+1)$ limit to radius R is identical to the radius-R limit, so the limiting family nests compatibly. Passing to the limit over $R = 1, 2, 3, \ldots$ via a diagonal argument then yields a well-defined consistent family of finite metric spaces that assembles into a pointed metric space (V, d, v). Convergence on every finite radius holds by construction. □

Example 13.7 (The k-regular tree as a pointed GH limit)**.** Let T_k be the k-regular tree with root v_0 and unit edge weights. Define G_n to be the finite induced subgraph of T_k on the ball of radius n about v_0, with $v_n := v_0$ and weights inherited from T_k. Each G_n is a finite rooted relational system.

The sequence (G_n, w_n, v_n) satisfies all three hypotheses of Theorem 13.6: the tree is k-regular (condition (i)), all edges have weight 1 (condition (ii)), and for any fixed R, the ball $B_{d_n}(v_n, R)$ stabilizes to $B_{T_k}(v_0, R)$ for all $n \geq R$ (condition (iii)). The limit is therefore (T_k, d, v_0), the full k-regular tree with its canonical metric. Tree-like (hyperbolic) geometry thus arises as a pointed Gromov–Hausdorff limit of finite relational systems, even though it is not compact.

13.3 The Compatibility Principle

Remark 13.8 (Parameterization, not approximation)*.* Theorem 13.2 demonstrates that classical spaces are compatible with the relational framework — they are recoverable from it via limiting constructions. However, the framework's position is that this compatibility should be understood as *parameterization* rather than *approximation*.

The sequence $(\mathbb{Z}/n\mathbb{Z})$ does not "approach" S^1 in the sense that S^1 is the "true" circle that the discrete systems are trying to become. Rather, each $\mathbb{Z}/n\mathbb{Z}$ is the circle at resolution n — a complete, exact geometric object in its own right. The Gromov–Hausdorff convergence records a relationship between those objects and the classical space S^1 that arises when the continuum substrate is supplied as an external context.

The framework does not need the limit to define the circle. The limit is available for those who want to understand the relationship to classical geometry.

14 Multi-Object Configurations and the Emergence of Space

The preceding sections treat each relational system as producing a single geometric object. This section addresses what happens when two or more relational systems are placed in relation to each other. The central result is that no ambient space is needed: space itself emerges from the inter-system relations, and its geometry is determined entirely by those relations.

14.1 Connecting Two Relational Systems

Definition 14.1 (Inter-system relations)**.** Let (G_A, w_A) and (G_B, w_B) be two relational systems with disjoint vertex sets V_A and V_B. An *inter-system relational structure* is a weighted directed graph G_{AB} with:

- edge set $E_{AB} \subseteq V_A \times V_B \cup V_B \times V_A$,
- weight function $w_{AB} : E_{AB} \to [0, \infty)$.

The *combined relational system* is $(G_A \sqcup G_{AB} \sqcup G_B,\ w_A \sqcup w_{AB} \sqcup w_B)$, where $\sqcup$ denotes disjoint union of graphs and weight functions.

Definition 14.2 (Combined metric)**.** Let d_A, d_B, and d_{AB} denote the induced metrics within G_A, within G_B, and the crossing distances via G_{AB} respectively. The *combined metric* on $V_A \cup V_B$ is the induced metric of the combined relational system:

$$d(x, y) = \inf_{\gamma} w(\gamma),$$

where the infimum is over all configurations γ in the combined system connecting x to y.

The combined metric recovers d_A within V_A and d_B within V_B, and introduces new finite distances between V_A and V_B whenever E_{AB} is non-empty. When E_{AB} is empty, all inter-system distances are infinite and the combined space is the disjoint union of two metrically disconnected objects.

14.2 Space as Emergent Relational Structure

The combined metric space $(V_A \cup V_B, d)$ is itself a finite metric space, and by Theorem 7.1 it arises from a relational system. It is not placed in a pre-existing ambient space — it *is* the space, produced entirely by the combined relational structure.

Theorem 14.3 (Space emergence)**.** *Let (G_A, w_A) and (G_B, w_B) be finite relational systems and let G_{AB} be an inter-system relational structure connecting them. Then:*

(i) The combined system induces a finite metric space $(V_A \cup V_B, d)$ in which the geometry of the combined space is fully determined by G_A, G_B, and G_{AB}.

(ii) No ambient space, background metric, or pre-existing coordinate system is required or assumed.

(iii) Different choices of G_{AB} over the same G_A and G_B produce genuinely different combined geometries.

Proof. (i) The combined system is a finite relational system; by Theorem 6.3 it induces a metric on $V_A \cup V_B$. (ii) No structure beyond G_A, G_B, and G_{AB} is invoked. (iii) Different choices of E_{AB} and w_{AB} produce different path sets between V_A and V_B, hence different infima and different combined metrics. □

Remark 14.4 (Position as inter-system relation)*.* The position of one object relative to another is not a property of either object. It is encoded in G_{AB}: the weights of the connecting relations determine how far apart the two objects are, and the structure of the connections determines the geometric relationship between them. Moving one object relative to another means changing G_{AB}, not changing either G_A or G_B.

This is the formal realization of the framework's claim that position is an external contextual parameter. In the single-object case that claim was stated but could not be demonstrated structurally. In the multi-object case it is a theorem: position lives in the relations between objects, not in the objects themselves.

14.3 Geometry of the Combined Space

The geometry of $(V_A \cup V_B, d)$ depends on the structure of G_{AB}. The following cases illustrate the range of possibilities.

Disconnected union. If $E_{AB} = \emptyset$, all distances between V_A and V_B are infinite. The combined space is the disjoint union of two metric spaces with no spatial relationship.

Single bridge. If E_{AB} consists of a single edge $e : v_A \to v_B$ with weight w, all paths from V_A to V_B pass through v_A and v_B. The combined space has a dumbbell geometry: two internal structures connected by a single corridor of length w.

Gluing along a subspace. If G_{AB} identifies a substructure of G_A with a substructure of G_B — assigns zero weight to connections between corresponding elements — the combined space is the metric gluing of the two systems along that common substructure. This is the relational analogue of the topological pushout: two spaces joined along a shared subspace. Unlike the continuous pushout of classical topology, this identification is a discrete finite operation: zero-weight inter-system relations collapse the two substructures into the same equivalence class under the induced pseudometric, with no limiting process or ambient space required. This is Commitment 4 applied to topological operations: even gluing is just a specific configuration of weights.

Product-like structure. If G_{AB} is a complete bipartite graph with uniform weights, and the crossing cost decouples from the internal metrics, the combined space approximates a product metric. Two discrete circles connected in this way produce a structure that, under refinement, approximates the flat torus $S^1 \times S^1$. This recovers the torus of Section 8.3 as a special case of two-object configuration.

Warped product. If the weights in G_{AB} vary depending on position within G_A — some vertices of A are closer to B than others — the combined metric is a discrete warped product. This is the relational analogue of the warped product metrics that appear in cosmological spacetime models.

Remark 14.5 (Resolution independence). The two systems G_A and G_B need not have the same resolution. A circle at resolution $n_A = 6$ and a circle at resolution $n_B = 600$ are the same relational object. When they are connected by G_{AB}, the combined metric is computed over the full combined graph. The crossing weights in G_{AB} operate at whatever scale is specified, independently of the internal scales of either system. The combined metric is sensitive to both the internal structure of each system and the scale of the connecting relations, but the two resolutions do not need to be commensurate. No synchronization of internal parameters is required for two objects to be placed in spatial relation to each other.

14.4 Space as the Invariant of a Configuration

The key observation of this section is that the combined metric space — the "space" containing both objects — is not a background structure into which the objects are placed. It is the invariant content of the combined relational configuration. Different configurations of the same two objects produce different spaces. There is no space prior to the configuration that determines where the objects sit. The space and the configuration are the same thing described from two perspectives: the relational perspective gives generators and laws; the geometric perspective gives the induced metric on the combined quotient.

This completes the framework's treatment of position. Position is not a property of an object, nor an external parameter imposed by a separate coordinate system. It is encoded in the inter-system relations, and the geometry of the resulting space is determined entirely by those relations together with the internal structure of each object.

15 Conclusion

The central philosophical shift of this framework is that space and geometric objects are not primitive entities but emergent structures arising from finite relational processes. We have presented a process-relational foundation for geometry in which:

- relations and compositions are the only primitives,
- configurations are finite processes over those relations,
- objects are equivalence classes under compositional congruence,
- all constructions are strictly finite and discrete.

The main theorems establish the core mathematical content. Theorem 5.2 guarantees that any congruence yields a well-defined quotient with associative composition. Theorem 6.3 shows that a metric arises from path optimization over weighted relations, with a discrete corollary (Corollary 6.4) when R is finite and weights are integer-valued. Theorem 7.1 establishes that every finite metric space arises from a relational system, making the framework universal for finite metric geometry. Theorem 11.10 shows that, after identifying metrically equivalent relational systems, the metric realization functor is an equivalence of categories between relational

systems and finite metric spaces. Theorem 13.2 establishes compatibility with classical continuous geometry via Gromov–Hausdorff convergence, understood as a bridge result rather than a foundational dependency. Theorem 13.6 extends this to non-compact spaces including the k-regular tree, using the pointed Gromov–Hausdorff framework. Section 14 establishes that space itself is not a primitive container but emerges from inter-system relations: placing two relational systems in relation to each other produces a combined metric space whose geometry is fully determined by the internal structure of each system and the connecting relations between them.

The examples demonstrate that familiar geometric structures—the circle, torus, tree, triangulated sphere, and integer grid—have clean discrete relational representations covering negative, zero, and positive curvature regimes, all without continuous or infinite machinery.

Taken together, the results establish a unified relational ontology for geometry. Distance is an ambient comparison between a relational object and a weight function at their interface. Position is an ambient comparison between two relational objects and the inter-system relations connecting them. Neither distance nor position is intrinsic to any geometric object, and neither requires an observer or external agent or a pre-existing container to be well-defined. Both are structural properties of the contact between mathematical objects — produced by that contact, belonging to neither party separately, requiring nothing beyond the relational data itself.

Limitations

We acknowledge several limitations of the present framework. First, it is not known whether all classical geometric and topological structures can be expressed as quotients of finite relational systems; this question is formalized as a future direction below. In particular, smooth manifolds, Riemannian metrics, and fiber bundles are structures whose representability within the framework remains open: it is unclear whether these can be faithfully encoded as finite relational quotients or whether they require structures that exceed the framework's expressive capacity. Second, continuous symmetry groups—Lie groups, the Poincaré group, $\mathrm{SO}(n)$—do not arise in the framework, as discussed in Remark 8.15; discrete cyclic approximations are available at any resolution but the continuous groups themselves are not objects of the theory. Third, the framework handles static geometric structures; extending it to dynamical settings where generators and congruences evolve over time is open. Fourth, while individual relational systems are computationally tractable, the complexity of multi-object configurations grows with the number and density of inter-system relations: determining properties of a combined space — such as its diameter, connectivity, or whether two configurations produce isomorphic combined spaces — is bounded by the complexity of the combined graph, which can grow rapidly as more systems are linked. This is not a limitation of the framework's expressiveness but of the computational resources required to analyze large configurations; it is the relational counterpart of the well-known computational hardness of large-scale metric and graph problems. These limitations are consistent with the framework's foundational commitment to finite, discrete, computable structure: they mark the boundary between what the framework expresses directly and what requires continuous or infinite machinery or additional computational resources. In each case the limitation is a precise, identifiable gap rather than a vague incompleteness, and in each case it points toward a concrete research direction.

Directions for further development:

1. **Multi-object space theory.** The multi-object framework of Section 14 opens several immediate questions: classify the possible combined geometries that arise from two given relational systems under all possible inter-system structures; characterize the conditions under which the combined space is a product, a gluing, or a warped product; and determine when two multi-object configurations produce isomorphic combined spaces. The connection to topological pushouts and amalgamated products suggests that algebraic topology provides the natural tools.

2. **Classification of congruences.** For single-generator systems, the pair (n, k) of Proposition 5.10 is a complete isomorphism invariant. For multi-generator systems, characterize which congruences produce isomorphic quotients and give a complete set of invariants. The natural framework is the correspondence between congruences and normal subgroupoids of the free path groupoid on R, generalizing the subgroup $S \leq \mathbb{Z}/n\mathbb{Z}$ of the single-generator case.

3. **Reconstruction of discrete classical spaces.** Embed known discrete structures (graphs, simplicial complexes, finite topological spaces) into the framework as special cases.

4. **Higher categorical formalization.** Develop the ω-groupoid hierarchy of Section 10 in full detail, maintaining finiteness at every level.

5. **Observational equivalence.** Determine when two relational systems are observationally equivalent—that is, produce isomorphic quotients under all congruences—and find a complete set of invariants.

6. **Applications to discrete physics.** Formalize discrete spacetime models (causal sets, spin foams) within the framework and characterize their curvature and holonomy relationally. A prerequisite is resolving the discrete-to-continuous curvature convergence question identified in Remark 8.7: establishing conditions under which the commutator-based discrete curvature converges to classical curvature tensors under Gromov–Hausdorff refinement, potentially via connections to synthetic curvature conditions in Alexandrov geometry.

7. **Representability limits.** Identify which classical geometric and topological structures — if any — cannot be expressed as quotients of finite relational systems, and characterize the boundary between representable and non-representable structures precisely. Understanding this boundary is directly relevant to quantum gravity: if certain spacetime topologies require structures that exceed what finite relational systems can express, that is a substantive constraint on any discrete spacetime model, not merely a mathematical technicality.

8. **Dynamical geometry.** Extend the framework to handle relational systems whose generators and congruences evolve — that is, geometry as a process rather than a static structure. This would provide a relational account of geometric dynamics, relevant to both discrete physics and time-varying computational models. In particular, it would allow the framework to address questions of curvature change, topological transitions, and the evolution of relational structure over time — currently expressible only in continuous settings.

9. **Cohomological and gauge-theoretic program.** The triangle defect of Proposition 8.8 and the commutator-based curvature of Section 8.4 suggest a deeper algebraic structure: the length functional $\ell : \mathrm{Mor}(\mathcal{C}) \to \mathbb{R}$ is a 1-cochain on the configuration category, and its coboundary $\omega(\alpha, \beta) = \ell(\alpha) + \ell(\beta) - \ell(\alpha \cdot \beta)$ is a 2-cocycle that encodes curvature. Under this identification, the Bianchi identity becomes the automatic cocycle condition $d\omega = d(d\ell) = 0$, flat geometries correspond to the trivial cohomology class $[\omega] = 0 \in H^2(\mathcal{C}, \mathbb{R})$, and δ-hyperbolic geometries correspond to bounded cocycle classes $|[\omega]| \leq \delta$.

   Extending this to a group-valued connection $A : \mathrm{Mor}(\mathcal{C}) \to G$ produces a curvature 2-cocycle $F(\alpha, \beta) = A(\alpha)A(\beta)A(\alpha \cdot \beta)^{-1}$ with gauge symmetry $A \mapsto g \cdot A \cdot g^{-1}$, yielding a full cohomological classification $[F] \in H^2(\mathcal{C}, G)$. In a smooth scaling limit satisfying second-order control, the commutator holonomy $H_\varepsilon(v, w) = \exp(\varepsilon^2 \mathcal{R}(v, w) + o(\varepsilon^2))$ recovers the classical Riemann curvature tensor, and the discrete Einstein–Hilbert action $S_\varepsilon[A] = \sum_x \mu_\varepsilon(x) \sum_{(\alpha,\beta) \in \mathcal{P}_x} \frac{1}{\varepsilon^2} \omega_\varepsilon(\alpha, \beta)$ converges to $\int_M R\sqrt{|g|}\, d^n x$, with stationary points recovering the Einstein field equations under metric-compatibility and torsion-free constraints. This program — from relational composition to gauge geometry to gravity — is a natural long-term direction for the framework.

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